

A Visual Trace Map of Edinburgh Place Names in Literature

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Abstract

Existing digital tools for literary geography plot place names as points on a map, treating them as geographic data. The meaning a place name carries in a literary text, however, comes from its narrative context rather than its coordinates, and this dissertation asks whether that narrative structure can be made visible directly, using the LitLong Edinburgh database of 620 literary works (1687–2015), 2,135 place names, and 50,248 place-name mention records. Two research questions are addressed: RQ1 asks whether an author’s distribution of place-name mentions across the city forms a visually distinctive spatial “fingerprint,” and RQ2 asks whether the co-occurrence of place names within literary works reveals a narrative topology distinct from their geographic topology. Six interactive D3.js visualisation prototypes were designed and implemented across these two directions: a radar chart, a bar-code fingerprint, and small-multiple maps for RQ1, and a force-directed network, a linear connection diagram, and a metro-style map for RQ2, all evaluated through a user study with domain-expert and general-public participants. The corpus analysis underlying RQ2 shows that Leith and Princes Street, roughly 2.5km apart, are the most strongly co-occurring place pair in the corpus (28 shared works). Rebuilding the metro-style map’s line structure from measured co-occurrence rather than hand-drawn geography raises the proportion of adjacent stations sharing a real book from 22% to 88%, surfacing further instances of narrative structure that geographic proximity alone could not reveal. The user study ($n = 13$) supports a qualified yes for RQ1: matching all five unlabelled fingerprints back to their authors reached 69% accuracy for the radar chart against a 20% chance baseline, with the bar-code and small-multiples designs close to half correct, and the radar chart the most consistently distinguishable design across every measure used, though single-shape identification accuracy was more modest and varied considerably by design.

Research Ethics Approval

This project obtained approval from the Informatics Research Ethics committee.

Ethics application number: 446635

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The participants' information sheet and a consent form are included in the appendix.

Declaration

I declare that this thesis was composed by myself, that the work contained herein is my own except where explicitly stated otherwise in the text, and that this work has not been submitted for any other degree or professional qualification except as specified.

(Merritt Wang)

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Chapter 1

Introduction

1.1 Motivation

Edinburgh is among the most literarily documented cities in the world. The Palimpsest project at the University of Edinburgh systematically extracted place-name mentions from hundreds of literary works spanning three centuries, producing the LitLong Edinburgh database: 620 documents published between 1687 and 2015, 2,135 distinct place names, and 50,248 geo-referenced mention records [1]. The project made this corpus publicly accessible through a web interface that plots place mentions on an interactive geographic map.

Geographic maps are a natural starting point for literary data of this kind, but they carry an implicit assumption: that the meaning of a place name is determined by its location. That assumption holds up fine in geography, less so in literature. When Irvine Welsh names Leith in *Trainspotting*, the word is not primarily a set of coordinates north of Edinburgh city centre; it evokes a social world, a class position, a political history. When Walter Scott names the Canongate, he is invoking a specific stratum of Edinburgh's past. A geographic map can show where these places are, but it has nothing to say about when in the narrative they appear, how often they recur, which other places surround them, or whose authorial voice chose them.

Place names in literary text, in other words, carry data about narrative structure as much as about geographic space. Accordingly, the visualisation problem tackled here is less about rendering literary geography more accurately on a map, and more about representing narrative structure in a form that reveals patterns geographic tools cannot.

1.2 Problem Statement

The core question this dissertation addresses is: can place names in literary text be visualised as an independent narrative structure, rather than as a rendering of geographic information?

It divides into two sub-problems corresponding to two distinct dimensions of narrative structure. The first concerns *authorial spatial attention*: does each author's distribution of place-name mentions across Edinburgh constitute a distinctive pattern that can be recognised without prior knowledge of the author? The second concerns *narrative co-occurrence*: do places that appear together across literary works form a topology that differs systematically from their geographic arrangement?

Existing tools do not address either question. They do not separate literary space from geographic space. They do not support comparison of authorial spatial styles. And they do not reveal which places are narratively proximate, as opposed to geographically proximate.

1.3 Research Questions

Two formal research questions guide the project:

RQ1 (Author Spatial Fingerprint): Does each author's distribution of place-name mentions across Edinburgh's official neighbourhood sectors form a visually distinctive spatial signature, sufficient for users to identify authors from their fingerprint alone?

RQ2 (Narrative Topology): Does the co-occurrence of place names within literary works reveal a narrative topology that diverges meaningfully from the geographic topology of Edinburgh?

The success criterion for RQ1, established in discussion with the supervisor, is that different authors must produce visually distinct fingerprint shapes on a one-to-one basis: no two authors should produce shapes that are indistinguishable to a naive user. The success criterion for RQ2 is that the visualisation should surface cases where narrative proximity contradicts geographic proximity, i.e. places that are spatially distant yet repeatedly appear together in the same literary works.

1.4 Contributions

This dissertation makes the following contributions:

1. A novel interactive visualisation system for literary place-name data that represents narrative structure rather than geographic location, implemented in D3.js and delivered as self-contained, deployable HTML files.
2. Two complementary visualisation directions: author spatial fingerprints (three designs) and narrative topology networks (three designs), each driven by real data from the LitLong Edinburgh corpus.
3. A validated spatial classification of Edinburgh place names using official administrative boundary data from the City of Edinburgh Council, comprising 14 sectors derived from the Neighbourhood Partnership Areas and Natural Neighbourhoods datasets (Open Government Licence v3.0).
4. An empirical finding with implications for literary geography: among five representative Edinburgh authors, no single place name is mentioned by three or more authors, and only 18 are shared by any two. Each author constructs an almost entirely private imaginary Edinburgh.
5. A method for extracting narrative reading order from the LitLong database by treating the `start_word` field as a globally monotonic index, enabling position-normalised analysis of when and where in a narrative each place appears.
6. A user evaluation comparing domain expert and general public responses to the six visualisation designs, informing the final design selection.

1.5 Dissertation Overview

Chapter 2 situates the project within three bodies of literature: literary geography and geocriticism, existing visualisation approaches for literary corpora, and information visualisation principles for humanities data. Chapter 3 describes the data exploration process, the design decisions made for each of the six visualisation candidates, and the rationale for the choices made. Chapter 4 covers the technical implementation, from database setup through to the D3.js interactive prototypes. Chapter 5 describes the user study design and reports its results. Chapter 6 discusses the findings in relation to the research questions and reflects on the project's limitations. Chapter 7 concludes and outlines directions for future work.

Chapter 2

Literature Review

2.1 Literary Geography and Geocriticism

The relationship between literature and place has been a concern of literary scholarship since at least the late twentieth century, but the theoretical grounding for a spatially-oriented reading of fiction consolidated around what is now called the “spatial turn” in the humanities [9]. Bertrand Westphal’s geocriticism argues that literary places are not simple reflections of real geographies but active constructions: representations that reshape the reader’s understanding of space and that exist in dialogue with the physical world rather than merely depicting it [11]. On this account, the place names in a novel are not pointers to locations on a map but nodes in a narrative network.

Franco Moretti’s quantitative approach to literary geography offered a different perspective [5]. By mapping the settings of hundreds of European novels, Moretti demonstrated that literature does not distribute itself uniformly across geographic space: certain places are obsessively returned to, others systematically avoided. He coined the phrase “literary silences” for the latter, absences that turn out to be as informative as presences. Moretti’s maps are geographic, but his interpretive framing is already narrative: he reads them for patterns of attention and neglect, not spatial accuracy.

Both traditions converge on a point that motivates this project: place in fiction is a narrative construction rather than a geographic fact. The visualisation problem, then, is less about representing the geography of literary Edinburgh more faithfully, and more about making its narrative dimensions (frequency, authorial style, co-occurrence, temporal position) visible.

2.2 The LitLong Edinburgh Project

The empirical foundation of this project is the LitLong Edinburgh corpus, produced by the Palimpsest project at the University of Edinburgh [1]. Using natural language processing pipelines, the project extracted geo-referenced place-name mentions from a curated selection of literary works set in or concerning Edinburgh, spanning the period 1687 to 2015. The full database, accessed here via a PostgreSQL dump released in 2022, contains 620 documents, 2,135 distinct place names, 50,248 mention records, and 424 authors, across 29 literary genre classifications.

Several features of the database are particularly relevant to this project. The `api_sentence` table stores the complete source sentence for every mention, enabling retrieval of the exact literary context in which a place name appears. The `start_word` field in `api_locationmention` encodes a global word index that is strictly monotonically increasing within each document, providing a proxy for narrative position without requiring full text parsing. The `api_posmention` table contains over two million part-of-speech annotations; these could support future analysis of whether a place name appears in dialogue, narration, or description, a dimension left unexplored here.

The original LitLong web interface renders place mentions as dots on an interactive geographic map. This design, while intuitive and technically accomplished, makes the assumptions critiqued above: it positions geographic location as the primary axis of literary meaning. The present project treats the LitLong data as the raw material for an alternative representational strategy.

2.3 Visualising Place in Literary Corpora

The closest antecedent to the present project is the work of Stange and Dörk on visualising literary place mentions in novels set in Berlin and New York [8]. Their “Novel City Maps” approach generated fingerprint-like representations of each novel’s spatial distribution across the city, using street-map coordinates as the positional axis. The present work extends this in a significant direction, replacing geographic coordinates entirely with an abstract categorical axis (city sectors), so the fingerprint shape represents authorial spatial attention without the perceptual noise real-world topography introduces. The data drove this choice: as Section 3.2 reports, place names in the LitLong corpus are too sparsely shared across authors for individual place names to serve as meaningful axes.

Other digital literary geography projects have largely retained the geographic map as their primary interface. Literary Atlas, Google's Literary Trips, and various GIS-based academic projects all plot textual mentions against spatial coordinates. While these tools are valuable for exploring the geographic distribution of literary references, they cannot address questions about authorial style, narrative co-occurrence, or the divergence between spatial proximity and narrative proximity.

The information visualisation literature offers relevant precedents for the two directions pursued here. Shneiderman's small multiples principle [10] (presenting multiple instances of the same visual form side by side for comparison) underlies the Small Multiples design in Direction 1. The force-directed graph layout used in Direction 2 follows Fruchterman and Reingold's energy-minimisation algorithm [4]. The metro-style map adapts Beck's insight that functional topology, the traveller's need to know which line and which direction, can be more usefully represented than geographic accuracy [2].

2.4 Information Visualisation for Humanities Data

Shneiderman's information-seeking mantra, "overview first, zoom and filter, details on demand" [7], provides a structural principle for the interaction design in this project. All six visualisation prototypes follow this pattern: an overview of the full dataset (all authors, all co-occurrence pairs) is presented by default, with interactive mechanisms for filtering and zooming, and detail panels that surface place-specific and book-specific information on demand.

Drucker argues that humanistic data should be understood as *capta*, actively interpreted and constructed, rather than as objectively given [3]. Several design decisions documented in Chapter 3 bear this out: the choice of sectors as axes reflects an interpretive act (what counts as a meaningful spatial unit for literary analysis?), and the choice of document-level co-occurrence as the granularity for the topology network reflects a judgment about what constitutes meaningful narrative co-presence.

Tufte's principle of small multiples [10] provides the theoretical foundation for the side-by-side comparison panel in Direction 1, where five maps of the same spatial extent but different author-specific bubble distributions enable direct visual comparison of authorial spatial patterns.

2.5 Graph and Network Visualisation

The force-directed network visualisation in Direction 2 follows the Fruchterman-Reingold algorithm [4], which models graph layout as a physical simulation in which nodes repel each other and edges act as springs. The critical property of force-directed layouts for this project is that node positions are determined entirely by the topology of the co-occurrence network, not by geographic coordinates. A node's position in the final layout reflects its narrative connectivity, not its map location.

Nöllenburg and Wolff's work on algorithmic metro-map layout [6] establishes that schematic map drawing (placing nodes on an octilinear grid while preserving topological correctness and legibility) is a computationally hard problem in the general case. An initial prototype of the metro-style visualisation sidestepped this problem by hand-drawing eight lines grouped by geography; a later audit (Section 3.5.0.0.1) showed that this hand-drawn grouping was only weakly supported by the measured co-occurrence data. The metro-style visualisation was therefore rebuilt around a heuristic pipeline: community detection on the co-occurrence graph, a co-occurrence-weighted spring embedding, and a local angle-correction pass to approximate octilinearity. None of this solves Nöllenburg and Wolff's problem exactly, but it derives the line structure and station layout from data rather than manual judgment.

2.6 Gap and Contribution

The literature reviewed above identifies three gaps that this project addresses. First, while literary geography theory has established that place in fiction is a narrative construction, existing digital tools continue to render literary place names as geographic data. Second, no existing tool supports direct visual comparison of authorial spatial styles as abstract fingerprints. Third, no existing tool visualises the co-occurrence topology of literary place names as a structure distinct from geographic proximity. This project addresses all three gaps through a system of interactive visualisations grounded in real corpus data and evaluated through a user study.

Chapter 3

Design

3.1 Design Rationale

The project brief specified D3.js as the implementation technology, and this choice aligns with the requirements of the visualisation task. D3.js is web-native, requiring no installation or server infrastructure for end users; this matters for sharing prototypes with supervisors and study participants. It provides fine-grained control over visual encoding, enabling the custom interactive designs required by both research directions. It is the standard tool in digital humanities visualisation practice, ensuring that the implementation approach is reproducible by others in the field.

The choice between static and interactive visualisation was settled early in the project. The LitLong corpus contains 620 documents, 2,135 place names, and 424 authors. A static visualisation that attempted to show all of this simultaneously would be unreadable. Interactivity is necessary to support the “overview first, zoom and filter, details on demand” pattern [7] that allows a reader to form a general impression and then drill into specific authors, sectors, or co-occurrence pairs.

The overarching design philosophy is that narrative structure should take precedence over geographic accuracy. Where geographic coordinates are used (the Small Multiples view), they serve as a spatial anchor to help users orient themselves, not as a claim that geographic position determines literary meaning. All other designs deliberately abandon coordinates in favour of abstract encodings that foreground narrative properties: sector-based fingerprints for Direction 1, and co-occurrence networks for Direction 2.

3.2 Data Exploration and Key Findings

Before designing the visualisations, a series of exploratory analyses of the LitLong data were conducted to establish what the data could support. These analyses produced five findings that directly shaped the design decisions documented below.

Finding 1: Author spatial languages are highly individual. Among the five test authors selected for prototyping (Alexander McCall Smith, Irvine Welsh, John Gibson Lockhart, Walter Scott, and Robert Louis Stevenson), no single place name is mentioned by three or more authors, and only 18 place names are shared by any two. That near-total absence of overlap confirms the literary-geographic intuition that each author constructs a largely private imaginary Edinburgh, with a direct implication for visualisation design: individual place names cannot serve as comparable axes across authors, because most axes would be zero for most authors. It was this finding that motivated the decision to use city sectors as the radar chart axes instead.

Finding 2: Sector distributions cross-validate against literary knowledge. The sector-based analysis yields distributions that align with established literary-geographic knowledge for all five test authors: McCall Smith concentrates in New Town (43.6%), consistent with his Scotland Street series; Welsh concentrates in Leith (44.3%), consistent with *Trainspotting*; Scott concentrates in Old Town (32.0%), consistent with his historical fiction; and Stevenson shows a notable Pentlands component (17.0%), consistent with his documented connection to Swanston. Between them these give reasonable confidence in the sector classification methodology.

Finding 3: Co-occurrence granularity requires document-level analysis. Sentence-level co-occurrence analysis yields zero pairs, since no sentence in the LitLong corpus contains more than one location mention, and page-level co-occurrence yields up to 92 co-occurring places on a single page, producing graphs too dense to be readable. Document-level co-occurrence, counting pairs of places that appear in the same work, yields 403 pairs with weight ≥ 2 (where weight counts the number of distinct works in which both places appear), and it was this granularity that was selected as the appropriate level of analysis for the topology network.

Finding 4: Narrative topology diverges from geographic topology. The strongest co-occurring place pair is Leith and Princes Street, with a co-occurrence weight of 28: they appear together in 28 distinct literary works. Geographically the two are separated by approximately 2.5 kilometres, Leith a port district to the north and Princes Street the main commercial thoroughfare of the city centre, so their narrative proximity plainly

does not reflect their spatial proximity. This is the clearest empirical illustration of the central claim behind Research Question 2.

Finding 5: Walter Scott's broad geographic range creates a display problem.

Scott's works reference locations far outside the Edinburgh city centre, including Forth (latitude 56.05), Haddington (longitude -2.80), and Linlithgow (longitude -3.58). When these points are plotted against a common coordinate scale in the Small Multiples view, they compress the city-centre data for all five authors into an unreadable band. The decision was made to fix the display range to the Edinburgh core area (latitude 55.85–56.02, longitude -3.35 to -3.05) and exclude out-of-range points from the map panel, while retaining them in the sector-based analyses.

3.3 Spatial Classification: Neighbourhood Sectors

A key methodological decision in Direction 1 was how to define the spatial units (sectors) that serve as axes in the radar chart and as ordering categories in the bar-code view. Three requirements guided this decision: the sectors must be few enough to serve as legible visual axes; they must be grounded in an authoritative external source so that the classification can be justified independently of the visualisation; and they must carry cultural and historical meaning relevant to literary analysis.

The solution adopted is a two-level framework derived entirely from official City of Edinburgh Council data, accessed via the Council's Open Spatial Data Portal (Open Government Licence v3.0). The primary layer is the **Neighbourhood Partnership Areas** (NP Areas), a set of 12 administratively defined zones used by the Council and its statutory partners for service planning and community consultation. These boundaries are legally established and publicly documented.

One NP Area, City Centre, was subdivided using a second official dataset, the **Natural Neighbourhoods** layer. Natural Neighbourhoods are resident-defined neighbourhood boundaries, created in 2004 as part of a ward boundary review and updated through a public online consultation in 2014. The City Centre NP Area encompasses two zones that are historically and culturally distinct in the Edinburgh literary tradition: Old Town (the medieval city, characterised by the High Street, the Canongate, and the closes of the Royal Mile) and New Town (the Georgian planned city built to the north from the 1760s onwards). These two areas are jointly inscribed as a UNESCO World Heritage Site, and their literary characters differ substantially: Old Town is the setting for the historical fiction of Scott and the Gothic sensibility of Stevenson and Hogg; New

Town is the milieu of McCall Smith’s contemporary domestic fiction and Lockhart’s nineteenth-century biography. Conflating them into a single City Centre sector would erase the most significant literary-geographic distinction in the corpus. A third Natural Neighbourhood unit, Canongate, was also retained as a separate sector, as it forms the eastern end of the Royal Mile and carries distinct historical significance.

The resulting framework comprises **14 sectors**: 11 unchanged NP Areas (Almond, Craigmillar/Duddingston, Forth, Inverleith, Leith, Liberton/Gilmerton, Pentlands, Portobello/Craigmillar, South Central, South West, and Western Edinburgh) plus Old Town, New Town, and Canongate derived from Natural Neighbourhoods. Place-name assignment proceeds by point-in-polygon intersection using the Shapely library. Of the 2,135 place names, 1,480 (69%) fall precisely within a sector polygon; 423 (20%) lie in inter-polygon gaps within the Edinburgh bounding box and are assigned to the nearest sector; and 232 (11%) lie outside Edinburgh entirely and are excluded from sector-based analyses. The resulting classification is cited as: “Copyright City of Edinburgh Council, contains Ordnance Survey data © Crown copyright.”

3.4 Direction 1: Author Spatial Fingerprint

Three visualisation designs were developed and implemented for Direction 1.

Radar chart (primary design). The radar chart represents each author as a polygon across the 14 sector axes, where the radius at each axis encodes the proportion of that author’s total mentions that fall within the corresponding sector. The polygon shape is the author’s spatial fingerprint. The chart was scaled to the maximum value in the dataset (0.436, McCall Smith’s New Town proportion) rather than to 1.0, because fixing the axis to 100% leaves over 60% of the radial space empty and compresses the meaningful variation into a small central region.

The primary interaction is point-level: hovering over any vertex reveals the sector percentage and a ranked list of the places and books that contribute to it, directly addressing the supervisor’s feedback that the visualisation should provide “low-level information upon interaction.” A detail panel opens on click, listing the top place names and top books in the selected sector for the selected author; hovering over an individual place row within the panel additionally surfaces a real example sentence drawn from `api_sentence`, together with the book and author it is drawn from. The legend supports filtering by author (click to hide/show) and isolation (double-click to show only one author), enabling the user to compare any pair or subset of authors.

Bar-code fingerprint (secondary design). The bar-code view represents each author as a horizontal row of vertical bars, one bar per place name, ordered from left to right by sector (north to south). The height of each bar encodes the proportion of that author's total mentions attributable to the corresponding place.

The Y-axis scaling went through three iterations before reaching its final form. The first prototype used a logarithmic scale, following standard practice for long-tail frequency distributions, but supervisor feedback noted that logarithmic scales are difficult for non-specialist users to interpret intuitively, and that the shape of the distribution mattered more here than its absolute magnitude. The second prototype switched to a linear scale shared across all five authors, which ran into a different problem: Welsh's Leith alone accounts for 44.3% of his total mentions, substantially more than any single place for any other author, so on a shared axis scaled to that maximum, every other author's distribution was compressed into a band occupying less than a quarter of the chart height, and within-author variation became imperceptible. The final design resolves this with a per-author independent Y-axis, where each row is scaled to its own maximum, prioritising the visibility of within-author spatial structure (the *shape* of each author's distribution) over cross-author magnitude comparison. The trade-off, that bar heights can no longer be compared directly across rows, is made explicit by a warning note in the interface; a shared-axis mode scaled to the global maximum remains available via a toggle for when cross-author comparison is what's wanted.

The bar-code view provides three ordering modes (by sector, by frequency, alphabetical) and two scale modes (percentage, absolute mention count), all switchable via controls at the top of the page. A text search field highlights the bars corresponding to a searched place name across all authors simultaneously, and a sector legend allows individual sectors to be hidden, narrowing the display to a subset of the city. Clicking any bar opens a detail panel listing the books that mention the selected place and a representative example sentence, mirroring the sentence-level detail added to the radar chart.

Small multiples (tertiary design). The small multiples view renders five side-by-side maps using Leaflet.js with a CartoDB Positron basemap. Bubble size encodes mention frequency (log scale by default, switchable to linear); bubble colour identifies the author. An optional synchronised-zoom mode locks all five panels to the same viewport, enabling direct spatial comparison. Clicking any bubble opens a popup with the place name, mention count, sector, and an example sentence. A sector-legend filter

hides bubbles outside a selected set of sectors across all five panels simultaneously, and a “compare two authors” mode replaces the five-panel grid with two larger, independently zoomable maps selected via dropdown menus, for closer side-by-side inspection of a chosen pair.

The CartoDB Positron style was selected over standard OpenStreetMap tiles because its reduced visual complexity (lighter colours, fewer labels) allows the data bubbles to remain visually prominent. The coordinate range is fixed to the Edinburgh core area as described in Finding 5 above.

3.5 Direction 2: Narrative Topology Networks

Three visualisation designs were developed and implemented for Direction 2.

Force-directed network (primary design). The force-directed network renders the document-level co-occurrence structure as a node-link diagram. Each node represents a place name; node size encodes total mention frequency across all authors. Each edge represents a co-occurrence pair, with edge width encoding co-occurrence weight using the function $w' = (w/w_{\max})^2 \times 12$, reached after testing two earlier mappings. A linear mapping ($w' = w \times 0.3$) produced edges that were visually indistinguishable in width across the full weight range 1–28; a square-root mapping ($w' = \sqrt{w/w_{\max}} \times 8$) improved discrimination marginally but still left the strongest connections insufficiently prominent relative to weaker ones. The squared function finally used produces a clear visual hierarchy while still rendering weaker connections as visible, thin lines.

Node positions are determined entirely by the force simulation (Fruchterman-Reingold via D3’s `forceSimulation`), with no geographic component. A node’s position in the layout reflects its pattern of co-occurrence connections, not its map location. This is a deliberate design choice: the diagram is meant to show narrative topology, and importing geographic position would conflate the two.

An interactive weight-threshold slider (range 1–28) allows the user to progressively reveal weaker connections, from the sparsest high-confidence view (threshold 28, showing only the single strongest pair) to the densest view (threshold 1, showing all 403 pairs). Nodes are draggable, and the diagram supports scroll-to-zoom and pan. Hovering over a node reveals a tooltip listing the five places most strongly co-occurring with it; clicking a node opens a detail panel with the same co-occurrence ranking, the books that mention the place, and an example sentence. A toggle switches node size between the absolute-frequency encoding described above and a percentage-of-total

encoding, trading a compressed logarithmic scale for a more literal but more visually dominated-by-outliers area encoding.

Linear connection diagram (secondary design). The linear connection diagram arranges place names along a horizontal axis, ordered left to right by total mention frequency. Bézier arcs connect co-occurring pairs above the axis; arc height is proportional to the horizontal distance between the two connected nodes, and stroke width is proportional to co-occurrence weight. A weight-threshold slider (range 1–28) controls which pairs are shown, and the horizontal ordering can be switched between frequency, sector, and alphabetical. Clicking an arc opens a detail panel listing the books in which both connected places co-occur; hovering over a place reveals an example sentence. A search field highlights a named place across the diagram.

The choice of a linear rather than circular layout followed a specific piece of supervisor feedback: a circular arrangement (as in a chord diagram) implies that the structure being represented is cyclical or symmetrical, which is inappropriate for literary narratives, which have a definite beginning and end. The linear axis does not make this implication. The leftmost position (highest-frequency place) corresponds loosely to centrality in the corpus, and the arrangement from left to right can be read as a rough ordering by narrative salience.

3.5.0.0.1 Designs considered but not selected. Two Direction-1 candidates were explored but ultimately deprioritised. A colour map silhouette design used a simplified Edinburgh coastline with area shading proportional to mention frequency; initial data analysis showed that all five authors concentrate their mentions in the city centre, producing visually similar heatmaps with limited discriminability between authors. A chord diagram was explored for Direction 2 but replaced by the linear connection diagram following supervisor feedback that a circular arrangement implies cyclical or symmetrical structure, inappropriate for literary narratives with a definite beginning and end.

Two further directions considered during early design were prototyped but deferred. A *Reader-plot timeline* would represent the sequence of place-name appearances across a single text on a horizontal narrative axis, directly exploiting the `position_pct` field in the `mention_order` table; the data infrastructure for this direction is complete, and it is identified as future work in Chapter 7. A *literary silences map* would visualise Edinburgh places that never appear in the corpus, following Moretti’s concept of literary silences [5]; this direction was deferred on methodological grounds, as the LitLong

corpus's coverage is not comprehensive enough to treat absence as meaningful evidence.

Metro-style map (illustrative design). The metro-style map arranges place names as stations on a small number of named lines, following the visual language of the London Underground map. The London Tube map, designed by Harry Beck in 1933, replaced geographic accuracy with functional topology: it told travellers which line to take and where to change, rather than where stations were physically located. This project applies the same substitution to literary geography, with lines representing narrative corridors (clusters of places that co-occur strongly in the same works) rather than physical routes. Beck's map, however, still encoded a real network: the lines corresponded to the physical tracks trains actually ran on, and only the geometry was distorted for legibility. An initial prototype of the literary metro map lacked that property. Eight lines were defined manually along geographic and cultural axes (a northern coastal line, an Old Town line along the Royal Mile, and so on), and station order within each line followed geographic sequence rather than measured narrative connection.

An audit of this prototype against the document-level co-occurrence data (Finding 3, Section 3.2) found that only 15 of 67 adjacent same-line station pairs (22%) shared at least one book in common: the map's line structure was a geographic argument wearing the visual grammar of a narrative one. The design was rebuilt around the following pipeline to close this gap.

1. **Community detection.** Modularity-based community detection (`networkx.community.greedy`) is applied to the 57-place, 403-pair co-occurrence graph (edges with weight ≥ 3), so that line membership is determined by measured narrative connectivity rather than manual geographic grouping. Communities larger than 18 stations are recursively re-clustered on their own induced subgraph, so that one line does not absorb several unrelated sub-clusters. Places with no strong connection to any cluster are folded into whichever cluster they connect to most strongly in the unthresholded graph.
2. **Station ordering.** Within each line, stations are ordered by a nearest-neighbour chain over real edge weights, starting from the strongest internal pair and appending each subsequent station to whichever end of the growing chain it connects to most strongly. This guarantees, by construction, that stations adjacent on a line are backed by measured co-occurrence wherever the underlying graph permits it.
3. **Interchanges.** A pair of stations on different lines with co-occurrence weight

≥ 10 produces a genuine interchange: the station is added as a real stop on the second line, rather than being connected by an illustrative cross-reference. This threshold was chosen empirically to yield a legible number of interchanges (12 of 57 stations); a lower threshold (weight ≥ 6) produced 28 interchanges, judged too dense to read as a small set of meaningful hubs.

4. **Line naming.** Names are assigned automatically: if a single author accounts for at least 55% of the book-mention volume attributed to a line’s home stations, the line is named after that author (e.g. “Welsh’s Edinburgh”); otherwise it is named after its two highest-mention stations (e.g. “Leith & Forth Corridor”).
5. **Layout.** A single shared coordinate is computed for every station using a co-occurrence-weighted spring embedding (`networkx.spring_layout`), so that two lines sharing an interchange meet at one physical point rather than being drawn in separate lanes and cross-referenced. Continuous positions are snapped to an integer grid with a collision-avoidance pass; the one station with zero measured co-occurrence is anchored to its assigned line’s centroid rather than left to the spring layout’s arbitrary placement of unconnected nodes.
6. **Octilinear correction and label placement.** Each line segment between two grid points is checked against Beck’s angle convention ($0^\circ/45^\circ/90^\circ$); segments that do not already satisfy it are redrawn as a two-part “dog-leg” (a diagonal run followed by a straight run) rather than an arbitrary-angle diagonal. Station labels are placed by a greedy search over eight compass positions at two candidate distances from each station, choosing the first position whose estimated text bounding box does not overlap an already-placed label or station marker.

This pipeline reduces the map to five data-derived lines (rather than eight hand-drawn ones) and raises the proportion of adjacency backed by real co-occurrence from 22% to 88% (46 of 52 adjacent station pairs; per-line figures are reported in Appendix B.5). The community structure also surfaced two findings invisible in the hand-drawn version, discussed further in Section 6: the word “Leith” clusters with the historical Old Town corridor associated with Scott and Lockhart rather than with the Welsh-dominated periphery line, and a small five-station line (Castle Street, Lasswade, the University of Edinburgh, the Royal Society of Edinburgh, and St Giles) emerged as a coherent biographical corridor, corresponding to the places Lockhart’s *Life of*

Walter Scott associates with Scott’s own residences and professional life rather than with Scott’s fiction.

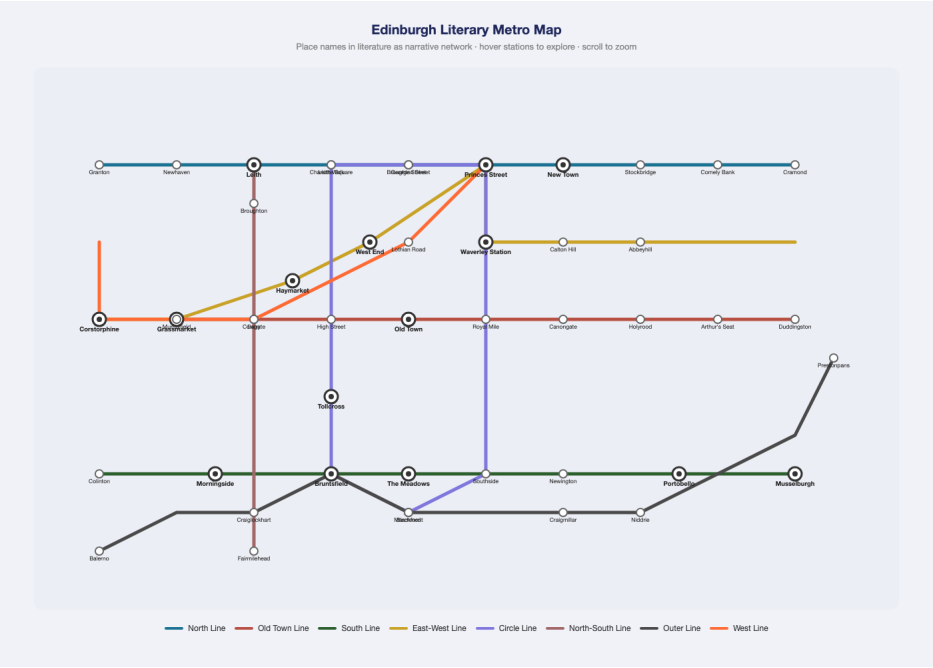


Figure 3.1: The initial hand-drawn metro-map prototype, before the redesign. Lines follow geographic and cultural axes (a North Line, an Old Town Line, an East-West Line) chosen by inspection, with only 22% of adjacent station pairs backed by measured co-occurrence.

3.6 Design Summary

Table 3.1 summarises the six visualisation designs and their key properties.

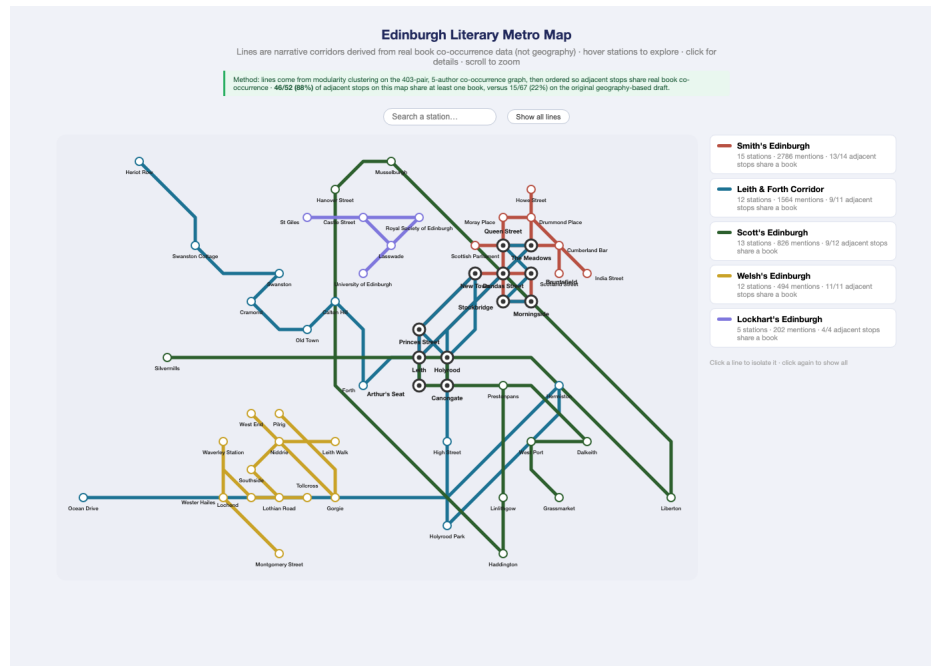


Figure 3.2: The same map after the redesign: five lines derived from modularity clustering on the co-occurrence graph and named after their dominant author, with 88% of adjacent station pairs now backed by measured co-occurrence. Compared with Figure 3.1, it is the line structure itself that has changed here, not just the layout.

Table 3.1: Summary of the six visualisation prototypes.

Direction	Design	Primary encoding	Key interaction
1	Radar chart	Polygon shape across 14 sector axes	Click vertex for sector detail panel; example sentence on hover
1	Bar-code	Bar height = mention share per place	Sector/frequency/alpha ordering; per-author Y-axis; search
1	Small multiples	Bubble size on real map	Sync zoom; log/linear toggle; sector filter; 2-author compare mode
2	Force-directed network	Node size = frequency; edge width = co-occurrence	Weight slider; drag nodes; click node for detail panel; absolute/percentage scale
2	Linear connection	Arc height = node distance; arc width = weight	Weight slider; sort by frequency/sector/alphabet; click arc for book list; search
2	Metro-style map	Line membership = community-detected narrative corridor	Click line to isolate; click station for detail panel; search

Chapter 4

Implementation

4.1 Data Pipeline

Database setup

PostgreSQL 16 was installed via Homebrew on macOS and a database named `litlong_edinburgh` was created. The full LitLong dump (`litlong_2022-06-21.sql`, 145 MB, approximately 2.3 million lines) was imported directly using `psql`. This dump is more complete than the four CSV snapshots provided with the project brief: it contains 620 documents (versus 548 in the CSV), 50,248 mention records (versus 48,056), and 27 tables not present in the CSV export, including `api_sentence` (full source text for every mention), `api_genre` (29 literary genre classifications), and `api_posmention` (2,085,834 part-of-speech annotation records).

One table failed to import from the dump: `api_location`, which uses PostGIS geometry types (`geom`, `poly` fields) that require the PostGIS extension. Since PostGIS was not installed, `api_location` was instead imported from the CSV snapshot using Python/pandas, retaining only the fields `id`, `text`, `lat`, `lon`, `gazref`, and `p_type`, and discarding the PostGIS-dependent geometry fields, which none of the current visualisation directions need since they use latitude/longitude centroids throughout.

The database name conflicts with user role assumptions in the dump (which was originally owned by user `palimpsest` on a remote server). Import warnings about missing roles are harmless and do not affect the data.

Narrative order extraction

The `api_locationmention` table contains a field `start_word` encoding a global word index for each mention (format: `w81489`). This field was validated to be strictly monotonically increasing within each document by extracting the integer values and checking for violations across a sample of documents. No violations were found. This property means that `start_word` can be used directly to reconstruct the reading order of place-name mentions within any document, without parsing the source text.

A custom table `mention_order` was constructed with the following fields: `id` (foreign key to `api_locationmention`), `document_id`, `location_id`, `word_num` (integer extracted from `start_word`), `mention_order` (rank of this mention within its document, 1 = earliest), and `position_pct` (normalised position within the document, computed as $(word_num - \min) / (\max - \min)$ per document). The `position_pct` field enables Reader-plot style timeline visualisations and will support future work on narrative sequence analysis.

Sector classification

The sector classification procedure is described in Section 3.3. The implementation uses the `shapely` Python library (version 2.0.1) for point-in-polygon intersection. The two GeoJSON boundary files were downloaded from the City of Edinburgh Council’s ArcGIS REST API:

- Neighbourhood Partnership Areas: Layer ID 28 at `edinburghcouncilmaps.info`
- Natural Neighbourhoods: Layer ID 188 at the same endpoint

Each place in `api_location` with a valid latitude and longitude was tested against the 14 sector polygons in order. If a point fell within a polygon, it was assigned to that sector (exact match). If it fell within the Edinburgh bounding box but outside all polygons, due to gaps between polygon boundaries, it was assigned to the nearest polygon by Euclidean distance in projected coordinates (nearest-neighbour assignment). Points outside the bounding box were assigned to a catch-all category, “Outer Scotland,” and excluded from sector-based analyses. The results were written to a derived table `location_sectors` and exported as a CSV for use in the D3 visualisations.

Co-occurrence computation

Document-level co-occurrence between place names was computed in Python using the `itertools.combinations` function. For each document, the set of distinct place names mentioned in that document was extracted; for every pair in that set, the pair's co-occurrence count was incremented by one. The resulting co-occurrence matrix was filtered to retain only pairs with weight ≥ 2 (appearing together in at least two documents), yielding 403 pairs. The top co-occurring pairs were exported as `network_edges.csv` and `network_nodes.csv`.

Sentence and book enrichment

A second pass over the same five-author mention data joins `api_sentence` and `api_document` to attach, for every place, a small sample of full source sentences (deduplicated, truncated to 220 characters, with the contributing book and author) and a ranked list of books that mention it. For Direction 2, the same join is additionally computed per co-occurrence edge, so the pair of places behind an arc or a network edge can be traced back to the specific books in which both appear. None of this alters any previously computed percentage, count, or co-occurrence weight; it is a separate lookup table, keyed by place name (and by place pair for edges), embedded alongside the existing data in each HTML file. Because sampling from `api_sentence` surfaces the source text verbatim, it can include informal register and strong language where the original does (for example, in Irvine Welsh's Scots-dialect prose). This is an authentic property of the corpus rather than a sampling artefact, but worth flagging before using these visualisations in a supervisor demonstration or a public-facing study session, since a hover tooltip is not the reader's first encounter with the surrounding literary context.

4.2 Direction 1: Author Spatial Fingerprints — Implementation

Radar chart

The radar chart is implemented using D3's `lineRadial` function with `curveLinearClosed`, which produces a closed polygon across the 14 angular positions. The angular spacing is $2\pi/14$ radians per sector. The radial scale maps sector proportion to a radius between

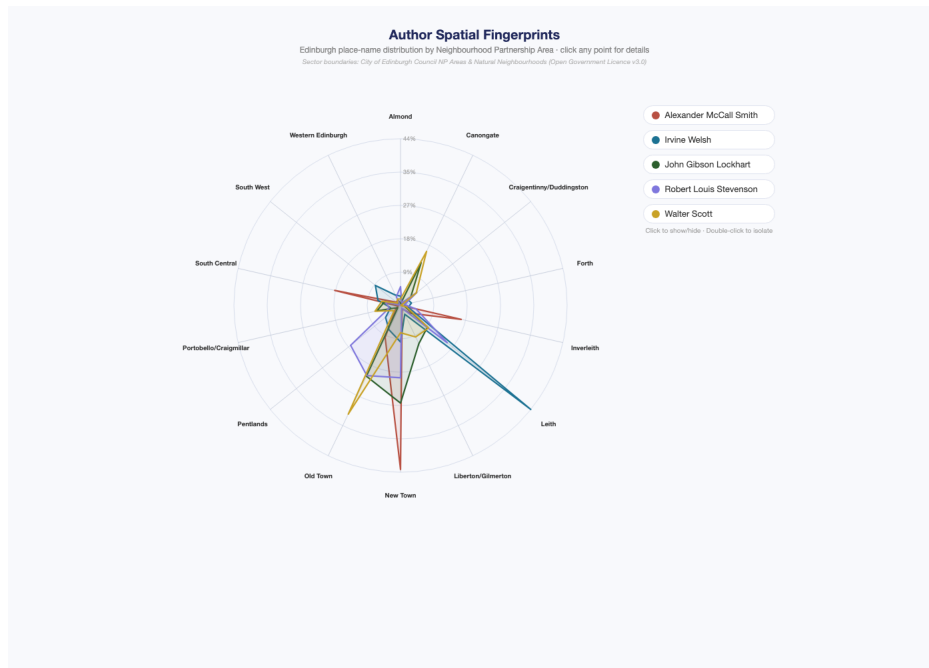


Figure 4.1: The radar chart (primary Direction 1 design), showing all five test authors simultaneously. Each polygon is one author's distribution of place-name mentions across the 14 city sectors; the near-total lack of overlap between polygons is the visual evidence for the “private imaginary geography” finding reported in Section 3.

0 and $R = 250$ pixels, where the scale maximum is set to the observed data maximum (0.436) rather than to 1.0.

Hover interaction is implemented using invisible `circle` elements of radius 14 pixels centred on each vertex, a value chosen after testing smaller ones: 6 pixels requires pixel-precise cursor placement and is difficult to target reliably, whereas at 14 pixels hovering is comfortable without the targets overlapping each other. On hover, a tooltip shows the sector name and percentage; on click, a detail panel opens showing the top eight place names in that sector for the selected author, ordered by mention count, together with a bar chart and the top five books.

Data is embedded directly as JavaScript arrays in the HTML file. This is necessary because browsers block `XMLHttpRequest` calls to local files under the `file://` protocol; a `d3.csv()` call would silently fail. The same embedding approach is used for all six D3 visualisations.

Bar-code fingerprint

Each author's row is rendered as an SVG element appended to a container div. The Y-axis domain is `[0, maxVal]`, where `maxVal` is the maximum proportion value for that author across all place names, not the global maximum across all authors. This per-author scaling is a deliberate design decision, as justified in Section 3.

Sector boundaries are visualised using two complementary encodings: a low-opacity background rectangle in the sector's colour spans the bars belonging to each sector, and a narrow colour strip below the bar area identifies the sector. Text labels are rendered below the colour strip only when the sector is wide enough to accommodate them without overlap (threshold: 40 pixels), with narrower sectors identified by colour alone, which avoids the text-stacking problem earlier versions had when all 14 sector labels were displayed regardless of available width.

Switching between ordering modes (sector, frequency, alphabetical) and scale modes (percentage, absolute) is implemented by re-rendering the SVG elements in JavaScript with the new ordering applied to the place-name array, rather than by animating existing elements. This approach is simpler to implement correctly and avoids the complexity of D3 transition management across multiple SVG groups.

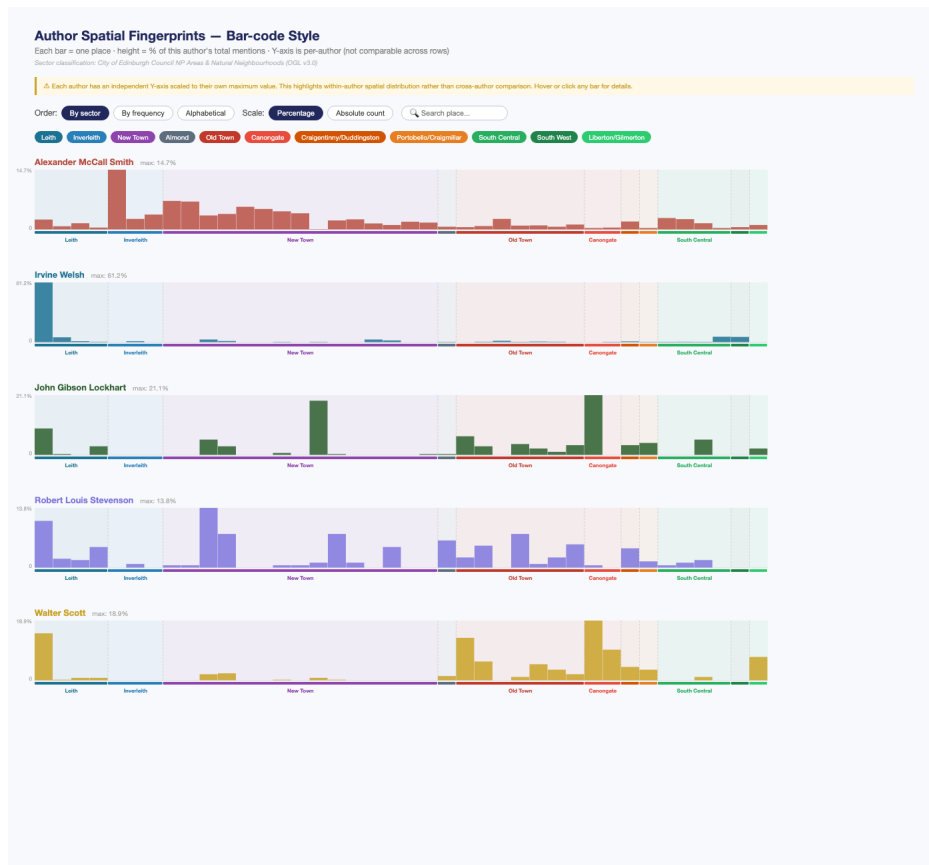


Figure 4.2: The bar-code fingerprint (secondary Direction 1 design). Each row is one author, each bar one place; bar height is normalised to that author's own maximum, so a short bar for a high-mention author is not visually confused with a tall bar for a low-mention author.

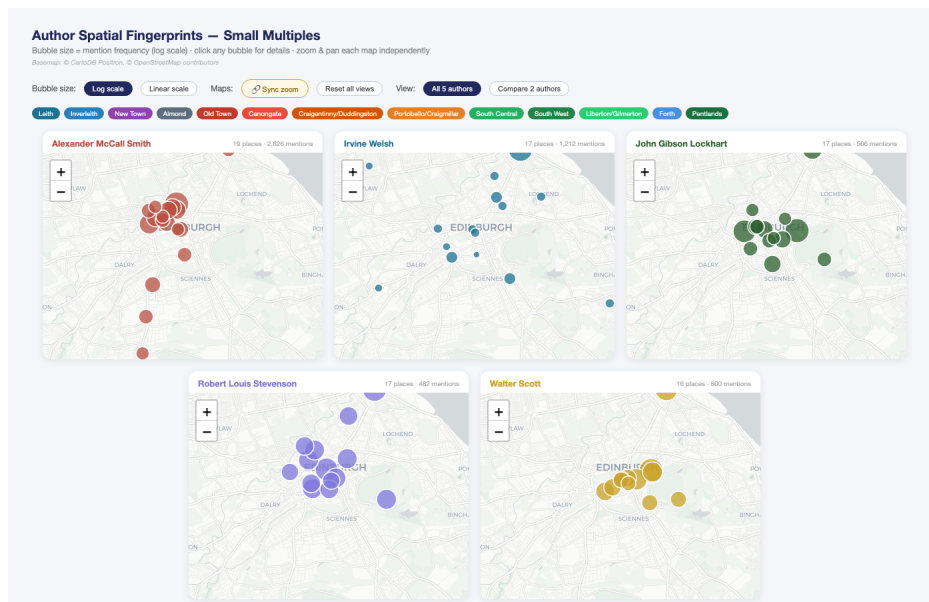


Figure 4.3: The small multiples design (tertiary Direction 1 design), five independent Leaflet maps side by side. Unlike the other two Direction 1 designs, this one retains real geographic coordinates as a spatial anchor, at the cost of the visual comparison being harder to make at a glance across panels.

Small multiples

Each of the five panels is implemented as an independent Leaflet.js map instance, initialised with the CartoDB Positron tile layer. Place-name bubbles are rendered as Leaflet `CircleMarker` objects, whose `radius` property is recomputed whenever the scale mode changes. Earlier prototypes used a two-stage Leaflet/D3 overlay, positioning D3 SVG elements over the map via screen-coordinate projection; this was replaced with native Leaflet `CircleMarker` objects, which handle map pan and zoom correctly without any manual coordinate projection.

Synchronised zoom is implemented by attaching a `moveend` event listener to each map; when the listener fires and the sync toggle is enabled, it calls `setView` on all other maps with the current centre and zoom level. A boolean `isSyncing` flag prevents the listener from triggering cascading updates.

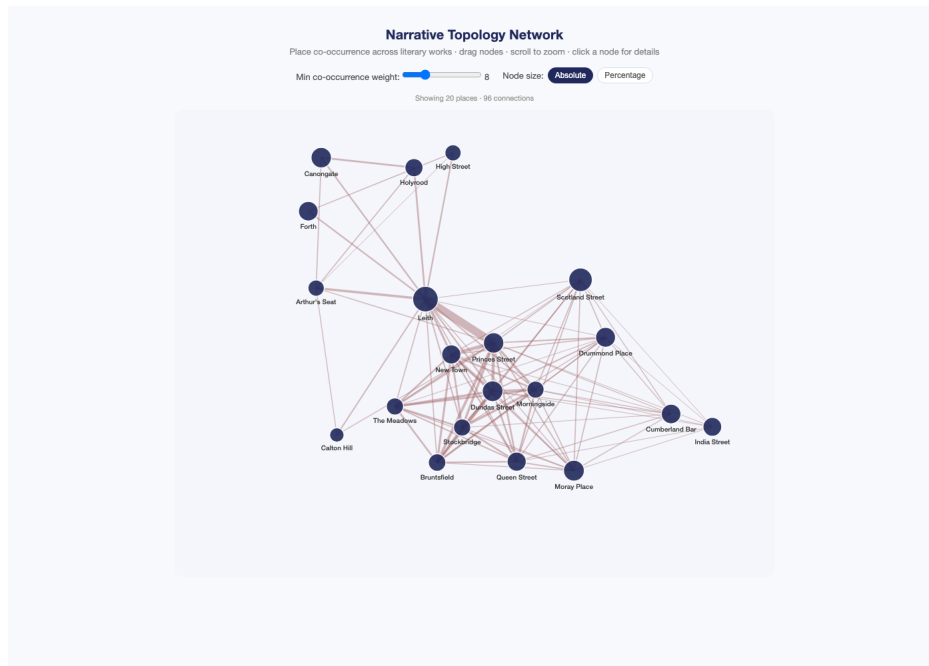


Figure 4.4: The force-directed network (primary Direction 2 design). Nodes are places, edges are document-level co-occurrence; node size reflects mention count. Leith and Princes Street, the strongest co-occurring pair in the corpus, are visibly pulled together despite the roughly 2.5km geographic distance between them.

4.3 Direction 2: Narrative Topology — Implementation

Force-directed network

The simulation uses four forces: `forceLink` (spring attraction between connected nodes, with link distance inversely proportional to co-occurrence weight), `forceManyBody` (uniform repulsion, strength -400), `forceCollide` (collision avoidance with radius equal to node radius plus 18 pixels), and `forceCenter` (gentle attraction to the canvas centre). The simulation is re-run from scratch whenever the weight threshold is changed, because the set of active nodes and edges changes and a smooth transition between graph topologies is not meaningful.

Node labels are positioned at $y = \text{node.y} + r + 12$ pixels, placing them below each node. An earlier version positioned labels at the node centroid with white fill, on the assumption the text would appear inside the node; in practice the labels were invisible, since the white text rendered against the equally dark node background. Moving labels below the node boundary fixed this.

Linear connection diagram

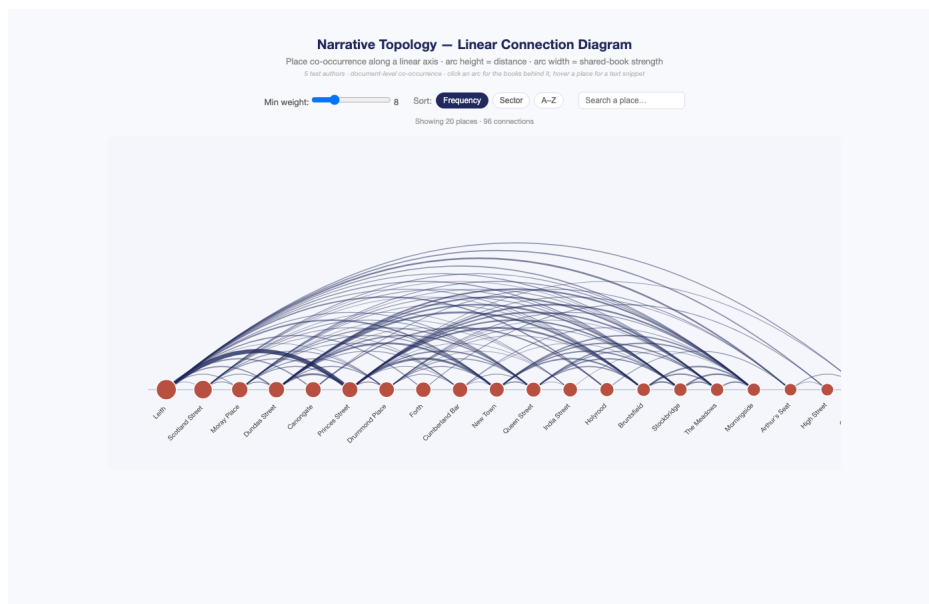


Figure 4.5: The linear connection diagram (secondary Direction 2 design). Places sit on a single horizontal axis; arc height and thickness both encode co-occurrence weight, so the strongest connections are immediately visible as the tallest arcs regardless of how far apart their endpoints are on the axis.

Each arc is drawn as a quadratic Bézier path: $M \ x1 \ y0 \ Q \ midX \ (y0 - h) \ x2 \ y0$, where $midX = (x1 + x2)/2$ and $h = |x2 - x1| \times 0.42$. The coefficient 0.42 was chosen to produce arcs that are visually distinct without extending beyond the SVG canvas for the longest connections. Stroke width is computed as $(w/w_{max})^{1.5} \times 6$.

Metro-style map

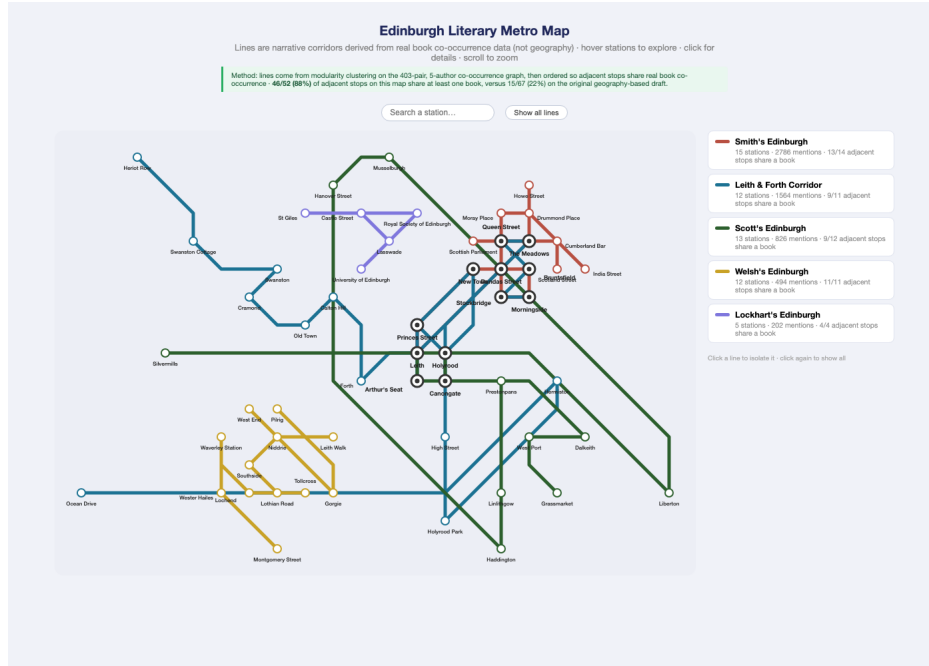


Figure 4.6: The metro-style map (illustrative Direction 2 design) after the data-driven redesign described in Section 3.5.0.0.1. Lines are named after their dominant author and derived from modularity clustering on the co-occurrence graph, not drawn by hand; interchange stations (double-ringed) mark places that multiple authors' narrative corridors both pass through.

The redesigned pipeline described in Section 3.5.0.0.1 is implemented as two Python scripts. `build_metro_lines.py` performs the community detection, station sequencing, interchange detection, and layout, and writes a single JSON file containing, per line, its name, colour, member stations, each station's shared (x,y) grid coordinate, and the adjacency-backed diagnostic count. `build_metro_html.py` then consumes this JSON and renders the final HTML file, embedding the data as a JavaScript object literal in the same manner as the other five visualisations. Keeping data derivation separate from rendering means the underlying clustering parameters (community-detection threshold,

interchange threshold, maximum line size) can be adjusted and the map regenerated without touching the rendering code at all.

Community detection is order-sensitive in an unwanted way: Python’s hash-randomised iteration over sets and dictionaries means that `greedy_modularity_communities` can break ties differently between runs, changing the order in which stations are sequenced along a line (though not which line a station is assigned to). Running the script with the environment variable `PYTHONHASHSEED=0` fixes this and was adopted as standard practice for regenerating the dataset.

At render time, each line is drawn as a single SVG path through its stations’ shared grid coordinates, converted to pixel coordinates and corrected for octilinearity (Section 3.5.0.0.1) by inserting an intermediate “dog-leg” point wherever two consecutive stations are not already aligned at a multiple of 45° . Because coordinates are shared across lines, two lines that both stop at the same interchange are drawn passing through the same physical point, producing genuine visual crossings rather than the separate parallel lanes used in the initial prototype. Interchange stations (those appearing on two or more lines) are marked with an outer white circle of radius 9 and an inner dark circle of radius 4, following the London Underground convention; all other stations use a smaller white circle with a coloured stroke matching their line’s colour. Station labels are positioned by the greedy eight-direction search described in Section 3.5.0.0.1; an initial fixed below-circle placement produced ten overlapping label pairs among the 57 stations in the densest cluster, reduced to two after adopting the greedy search. D3 zoom behaviour is attached to the SVG to support scroll-to-zoom and drag-to-pan; clicking a line or a legend entry isolates it by dimming all other lines and stations, and a text field highlights matching station names.

4.4 Technical Decisions and Constraints

Data embedding. All six D3 HTML files embed their data as JavaScript object literals. The trade-off is larger file sizes and the need to regenerate the HTML file when the underlying data changes. For a research prototype evaluated with a fixed dataset, this is an acceptable constraint.

Versioning. Each HTML file is versioned by maintaining previous versions under filenames such as `radar_v1.html`, `network_v1.html`, etc. This preserves the design evolution as a record for the dissertation and for the supervisor.

Browser compatibility. All files have been tested in Safari and Chrome on macOS.

D3.js v7 and Leaflet.js 1.9.4 are both loaded from CDN at runtime, requiring an internet connection.

PostGIS. The PostGIS extension was not installed, and the geometry fields of `api_location` (`geom`, `poly`) are excluded from all analyses. Latitude/longitude centroid coordinates are used throughout. For the current visualisation directions, centroid coordinates are sufficient; more precise polygon representations of place areas would be needed only for overlay analyses that are not part of the current scope.

Colour accessibility. The five fixed per-author colours were checked against simulated protanopia, deuteranopia, and tritanopia (the Brettel/Viénot transform applied to each hex value) rather than assumed to be distinguishable. All ten pairs remain visually separable under all three simulations; the closest pair in every case is McCall Smith and Scott, noticeably closer together than the other pairs though still above the threshold treated here as a likely confusion risk. This is a reasonable but not maximally optimised palette, since the colours were originally chosen for contrast against the interface background rather than for colour-vision-deficient legibility specifically.

4.5 Combined Interface

The combined interface (`data/processed/combined/d3/combined_interface.html`) implements the navigation shell described in the original plan and, unlike the first iteration, the author selector and cross-view linking are now functional rather than scaffolded. The shell provides top-level navigation between the two directions (Fingerprints / Topology), a left-hand panel switching between the three visualisations within the active direction plus the shared author selector, a main view, and a right-hand detail panel.

Five of the six visualisations (radar, bar-code, small multiples, network, linear) render live in-page from a single shared `selectedAuthors` array, reusing the same client-side top-15-place and document-level co-occurrence computation already built and validated for the scale exploration (Appendix B.6). The preset buttons (2 / 5 / 20 / 50 authors) and free-text search operate on the same four JSON bundles (`all_authors_radar.json`, `all_authors_barcode.json`, `all_authors_small_multiples.json`, `all_authors_network.json`), so no new data pipeline had to be built just for this. Cross-view linking is implemented as a single `focusedAuthor` value alongside the selection: clicking an author's polygon (radar), row (bar-code), or map panel (small multiples) toggles focus and re-renders that view with the focused author emphasised and the rest dimmed, and switching to a Topology view recomputes that author's own

top-15 places and highlights the matching nodes/edges gold within the current selection's combined graph, dimming the rest. That gives the planned behaviour ("selecting an author in fingerprint view highlights their places in topology view") without needing `postMessage` between iframes at all, since the five views now share one document rather than being embedded separately.

The metro map remains the one exception. Because its lines depend on offline community detection (Section 3.5.0.0.1) rather than a computation that can run per keystroke in the browser, it stays an `<iframe>` that switches between four precomputed snapshots (2, 5, 20, and 50 authors, chosen by nearest log-distance match to the current selection) rather than recomputing live, and it does not participate in the focus highlight or the detail panel below. An earlier version of this switch only had two snapshots and picked between them with a fixed threshold, so selecting 2 versus 5 authors, or 20 versus 50, produced no visible change at all. Three snapshots have since been added, at 2 and 50 authors (96% and 73% adjacency-backed respectively, extending the monotonic trend already reported for 5 and 20), so every preset now shows a genuinely different, correctly rendered map. Past roughly 20 authors, though, the snapshot is really showing community detection degrading rather than offering a usable exploration surface, so both the standalone scale-exploration tool and the combined interface display a prominent warning at that scale rather than a quiet note, to make clear this is a reported finding and not a rendering fault.

A unified outer detail panel, shared across the five live views, is also implemented, backed by a fifth dataset, `all_authors_sentences.json`. This is generated by a new query against `api_sentence` that mirrors the query already used for the five-author prototypes (same sampling, same 220-character truncation, same random seed), but drops the author filter and restricts each of the 408 authors to their own top-15 places by mention count, exactly the set reachable through the live selector, so no sentence data ends up generated for places a user could never hover over. Hovering a place in bar-code, small multiples, network, or linear looks up every author in the current selection against this dataset and merges their example sentences and mention-weighted book list for that place; hovering an author's shape in the radar chart shows their single most-mentioned place as a representative example instead, since radar's own vertices are sectors rather than individual places. The panel is deliberately sticky, not clearing on mouse-out, so a sentence stays readable after the cursor moves away; it does clear when the active visualisation or the author selection changes, since otherwise a stale sentence could reference a place or author no longer in view.

Selecting a large author subset in network or linear turned out to produce enough co-occurrence edges (12,833 at the 50-author preset) that d3's force simulation failed to settle within several seconds, leaving the interface visibly unresponsive. The weight-threshold slider compounded this: it was capped at a fixed maximum of 10 regardless of the true maximum edge weight in the current selection (which reaches into the 80s at scale), so it could not be used to filter down far enough to recover. Both views now cap rendering to the 1,500 strongest-weighted edges, with the slider's maximum computed dynamically from the current graph, and the underlying author selection is left untouched by the cap so switching to another view still reflects the full selection. A related legibility issue turned up in the radar chart's per-author colour assignment: a fixed-step hue formula placed two of fifty authors' colours one degree apart on the colour wheel, indistinguishable despite sitting on the same chart. Hues are now spaced evenly across the current selection size instead, guaranteeing a minimum perceptual gap between every pair shown.

A supervision meeting reviewing the interface at this stage produced three further revisions.

First, the interface had accumulated implementation-facing language visible to anyone using it: the top banner named `api_sentence` and internal rendering limits, and small badges read "live"/"implemented" next to section headings. None of that helps someone exploring the data decide what to do next, so it was rewritten throughout in plain terms (the metro map's note, for example, now reads that it is "pre-drawn rather than redrawn live" instead of describing offline community detection).

Second, the radar chart's hover interaction turned out to offer only one tooltip per author's entire polygon, with no way to inspect a single sector. A small marker was added at each of the 14 sector vertices; hovering one now shows that author's mention count and percentage for that specific sector, together with the individual place names contributing to it (cross-referenced from the bar-code dataset), and updates the outer detail panel to that sector's leading place. The small multiples map markers were extended the same way, gaining sector and percentage-of-total alongside their existing place and mention count.

Third, author selection had previously offered only fixed presets plus one-at-a-time typed search, which does not make "pick any combination of the 408 authors" obvious as something you can even do. A collapsible, scrollable checklist of all 408 authors, sorted by mention count, was added alongside the existing controls, and a one-line note now clarifies that the "2"/"5" presets are the project's fixed core-author set while

“Top 20”/“Top 50” means ranked by total corpus mentions, rather than leaving “top” unexplained.

The metro map’s own further redesign was also discussed in this meeting; the supervisor’s response to the current version was positive, and any next iteration is deferred pending a separate follow-up discussion rather than being scoped here.

A fifth author preset, loading all 408 located authors at once, was added alongside the existing 2/5/20/50 presets and explicitly labelled “large-scale exploration” in both its button text and an accompanying note. This distinguishes it from the four practical presets: it is meant for spotting broad, corpus-wide patterns rather than reading individual detail, and it is the rendering caps already described above for network, linear, small multiples, and metro that keep it usable at this scale rather than anything that needed disabling.

Chapter 5

Evaluation

5.1 Study Design

The user study is designed to evaluate whether the six visualisation prototypes support the tasks implied by the two research questions: author identification (RQ1) and narrative-relationship identification (RQ2).

Participant groups. Two groups are recruited: domain experts (literary scholars, digital humanists, and researchers with familiarity with Edinburgh’s literary geography) and general public (participants without specialist knowledge of Edinburgh literature or information visualisation). Recruiting two groups allows the study to test the hypothesis that expert and non-expert users differ in which design they find most legible, and in whether they spontaneously perceive the divergence between narrative and geographic proximity.

Format. The study is conducted online to maximise recruitment reach and sample size. An online format also ensures that participants interact with the visualisations in a browser environment that reflects real-world usage conditions.

Introduction protocol. Participants are shown each visualisation with only a brief description of the project’s purpose (“We are studying how literary place names can be visualised”), but no explanation of what each visual encoding means or how to read it. This “cold” presentation matters methodologically, since it tests whether the designs are self-explanatory rather than testing participants’ ability to apply an explanation they were given: a participant who needs the design explained before they can use it is, in effect, reporting a usability problem. This is distinct from the labelled reference view used in Task 1 below, which withholds no encoding explanation either; it simply gives participants something to look at before being tested, rather than telling them what to

look for.

Tasks. Two task types are used. Task 1 (fingerprint direction) tests the success criterion for RQ1 set out in Chapter 1: whether the five authors' fingerprint shapes are distinct enough to be learned and then recognised, not whether a participant can name an author from memory of their books with no visual reference at all – the latter would conflate visual distinctiveness with prior literary knowledge and would be close to impossible for a participant with no background in these five authors specifically, regardless of how good the encoding is. Task 1 is accordingly a two-step study-then-identify task. Participants first see a labelled reference view showing all five authors' fingerprints together for the design in question (the same canonical, all-five-authors prototype described in Chapter 3), and are asked to spend a few minutes noticing how the shapes differ, without being told what to look for. They then see a purpose-built single-author variant of the same design, showing one author's shape only, with the legend and the click-through detail panel (which would otherwise expose book titles and could reveal identity through bibliographic or authorial-voice recognition) removed entirely, leaving only the shape and its hover-revealed sector values, and are asked which of the five authors they now believe it represents, with the reference view still available to revisit. The single-author variant is rendered in a neutral colour rather than that author's colour from the reference view; an earlier version of this stimulus reused the same per-author colour in both views, which would have let a participant answer correctly from a remembered colour swatch alone, without the shape itself playing any role – testing colour recall rather than the pattern-learnability this task is meant to measure. Removing the colour cue does not by itself rule out a related failure mode: a participant could still answer correctly by matching the shape's outline against a memorised image from the reference view, without engaging at all with what that outline actually communicates – which geographic sector or place the author's writing concentrates on. Task 1 accordingly opens with an objective, answer-keyed question that can only be answered from the blind stimulus itself, with no memory of the reference view required, asking which sector or place the shape indicates the author writes about most; a participant who cannot answer this correctly has not read the encoding, regardless of whether they go on to guess the right author from shape recognition alone. Accuracy on this identification is the primary metric, but the open-text question asking what specific feature of the pattern led to that answer is treated as equally important, since it captures whether participants can articulate a distinguishing feature even when the forced-choice answer is wrong. A different author is used for

each of the three designs, so that a participant's answer on the first fingerprint task does not anchor their answer on the second and third. A single blind identification, however, only ever yields a correct/incorrect judgement against a one-in-five baseline, which says little about which specific authors' fingerprints are most easily confused with one another. Task 1 therefore adds a third step: participants see all five authors' shapes again, unlabelled and in a shuffled order, and match each shape to an author, producing a full 5×5 confusion matrix per design rather than a single accuracy figure, plus a self-rated judgement of how easily the five patterns could be confused overall. Both task directions also include items added specifically to guard against ceiling and floor effects on the primary forced-choice and free-text questions: Task 1 adds a runner-up guess after the initial identification and a two-alternative forced choice between the single most confusable pair of authors for each design, identified in advance by cosine similarity on the underlying data; Task 2 adds a rating of three place pairs of known strong/medium/weak co-occurrence weight and a forced choice between a genuine data-derived cluster and two constructed distractors. Full wording and the data-driven basis for each choice are given in Appendix A.3. Task 2 (topology direction): participants are shown a topology visualisation (force-directed network, linear connection diagram, or metro map) and asked to identify the two places that appear most strongly connected, and to judge whether those two places are geographically close or far apart in Edinburgh. These three designs already aggregate all five authors' co-occurrence data into one structure with every place labelled throughout, so there is no identity to guess and no equivalent reference-then-test structure is needed. The secondary question tests whether the visualisation spontaneously communicates the narrative/geographic divergence finding. After completing all tasks, participants vote for the most intuitive design in each direction.

Ethics. The study has received ethics approval from the Informatics Research Ethics Committee (application number 446635). Participants complete an online consent form before beginning the study. The participant information sheet and consent form are included in Appendix A.

5.2 Participants

Sixteen responses were collected; three were excluded before analysis on data-quality grounds (Section 5.3), leaving $n = 13$. Participants were recruited through the researcher's personal and academic network within the UK, using the recruitment chan-

nels and eligibility criteria set out in the approved ethics application; recruitment was not restricted to the University of Edinburgh.

Grouping into “domain expert” and “general public” was based on participants’ self-described background (Q-B3) rather than assigned at recruitment, following the design set out in Section 5.1. Four participants reported a background in literary studies/English literature, digital humanities, or information visualisation/HCI/computer science and were treated as the expert group; the remaining nine, spanning other academic backgrounds and no specialist background, formed the general-public group. This four-to-nine split is thinner on the expert side than the five-to-eight target set during recruitment planning, and is treated throughout this chapter as an exploratory rather than a statistically powered comparison.

A striking feature of the sample, expert group included, is near-uniform unfamiliarity with the five specific authors under study: nine of the thirteen participants had read none of Alexander McCall Smith, Irvine Welsh, John Gibson Lockhart, Walter Scott, or Robert Louis Stevenson, and eight rated themselves “not at all familiar” with Edinburgh’s literary history. Even the one participant with a literary-studies background reported being only “slightly familiar” and had read none of the five authors. This is not a weakness of recruitment so much as a confirmation that the identification tasks in Section 5.4 are testing what they are meant to test: whether the visual encoding itself is learnable, rather than whether a participant already happened to know these particular authors. General familiarity with reading data visualisations was more evenly spread, from “slightly familiar” to “extremely familiar,” with “moderately familiar” the most common response; four participants reported having lived in Edinburgh, one had visited once or twice, and eight had never visited.

5.3 Study Procedure

The study was distributed as an anonymous Qualtrics link and completed online, unmoderated, and at each participant’s own pace, following the design set out in Section 5.1. Median completion time among the thirteen retained responses was 46 minutes, in line with the estimated 50–65 minutes; the recorded range was wider than this because Qualtrics’ “finish later” option, enabled given the survey’s length, let a handful of participants leave the survey open across more than one sitting, which inflates elapsed time without reflecting active engagement.

Three of the sixteen raw responses were excluded before analysis. One was com-

pleted in under ten minutes with near-empty free-text answers (e.g. a single full stop in place of an explanation); a second gave single-letter or otherwise non-committal answers (“a”, “b”, “no”) on every free-text field, including ones the anti-memorisation questions (F0, T1b) specifically ask participants to justify in their own words; a third was completed in six minutes with free-text answers unrelated to the questions asked. All three were judged, on this basis, not to reflect genuine engagement with the task, and were removed prior to any quantitative analysis; they are not included in the $n = 13$ reported throughout this chapter. A small number of free-text answers in the retained responses were originally given in Chinese by participants who completed the study while resident in the UK; these are translated by the researcher and reproduced alongside the original wording in the underlying response data, rather than replacing it.

5.4 Results

Fingerprint identification and matching (RQ1). Results below are reported per design, out of $n = 13$, with the single-author identity used for the blind stimulus given in parentheses.

Radar chart (Walter Scott). The encoding-legibility check (F0) was answered correctly by 10/13 participants (77%), the highest of the three designs. Single-shape identification (F1) was correct for 6/13 (46%), rising to 8/13 (62%) once a correct second guess (F1b) is credited. On the five-way matching task (M1), 45 of 65 shape-to-author assignments were correct (69%, against a 20% chance baseline), the highest matching accuracy of the three designs; the most common confusions were between Robert Louis Stevenson and Irvine Welsh, and between John Gibson Lockhart and Robert Louis Stevenson. On the targeted hardest-pair check (P1, Lockhart vs Scott), 11/13 (85%) correctly distinguished the two.

Bar-code (John Gibson Lockhart). F0 was correct for 6/13 (46%). F1 was correct for 6/13 (46%), rising to 10/13 (77%) on the Top-2 measure, the largest Top-1-to-Top-2 gain of the three designs, consistent with participants narrowing the shape to a plausible pair without landing on the exact author. M1 matching accuracy was 31/65 (48%); the most frequent confusions were between Robert Louis Stevenson and John Gibson Lockhart, and between Irvine Welsh and Walter Scott. P1 (Scott vs Welsh) was correct for 10/13 (77%).

Small multiples (Robert Louis Stevenson). This design produced the weakest results on every fingerprint measure: F0 correct for 5/13 (38%), F1 for 3/13 (23%), Top-2

for 5/13 (38%), and M1 matching accuracy 32/65 (49%), with Walter Scott the author most often confused for others. P1 (Lockhart vs Stevenson) was nonetheless correct for 10/13 (77%), comparable to the other two designs, suggesting that even where overall identification is weak, the single most-confusable pair is not disproportionately harder here than for the stronger designs.

Self-explanatoriness (F4, “would you have understood this without explanation”) was mixed for all three designs, with no design rated “yes, easily” by more than a third of participants. In the post-task ranking, the three fingerprint designs were close to an even three-way split (small multiples 5 votes, radar chart 4, bar-code 4).

Narrative topology (RQ2). All three topology designs share the same underlying co-occurrence data, so the same ground-truth pair (Leith and Princes Street) and design-specific place pairs/clusters documented in Appendix A apply throughout.

Force-directed network. T1 (identify the strongest pair) was correct for 9/13 (69%). Ratings on the three fixed strength-check pairs (T-Strength) correlated strongly with the pairs’ real co-occurrence weights ($r = 0.64$, $p < 0.001$, $n = 39$ ratings), the strongest correlation of the three designs. T-ClusterVerify was correct for 9/13 (69%), matching T1’s accuracy.

Linear connection diagram. T1 was also correct for 9/13 (69%), identical to the network. The strength-rating correlation was weaker and not conventionally significant ($r = 0.30$, $p = 0.067$, $n = 39$). T-ClusterVerify accuracy was markedly lower than the other two designs at 3/13 (23%); the large majority (8/13) instead selected a distractor grouping (New Town, Leith Walk, Holyrood, Waverley Station) built by deliberately mixing places from different real communities, suggesting the linear diagram’s arc-based encoding does not communicate cluster membership as clearly as the other two designs, despite matching them on the single strongest-pair task.

Metro-style map. T1 accuracy was substantially lower than the other two designs at 2/13 (15%), even though the same underlying pair is, by construction, the single strongest edge in every design. The strength-rating correlation was weak but statistically significant ($r = 0.34$, $p = 0.035$, $n = 39$). T-ClusterVerify accuracy was 7/13 (54%), between the other two designs. Despite this being the weakest design on two of the three objective measures, it was the clear winner of the post-task ranking (7/13 votes, against 5 for the network and 1 for the linear diagram), and of the final “most memorable visualisation” question (Q-R5) among those who named a specific design.

5.5 Discussion of Results

The clearest single result in this chapter is that the radar chart outperforms the other two fingerprint designs on every measure computed: the highest F0 comprehension rate, the highest single-identification and Top-2 accuracy, the highest five-way matching accuracy, and the highest hardest-pair discrimination. This is consistent with the qualitative pattern in participants' F3 answers, where several radar-chart responses named a specific sector by name ("a very large spike in Leith area", "the maximum is old town"), whereas bar-code and small-multiples answers more often described a feature without naming what it corresponded to ("peak value", "an unusually large spike"). The polygon's asymmetry appears to give participants a concrete verbal handle on the pattern that the other two encodings, for this sample, did not as reliably provide. Given that F0 accuracy tracks F1 accuracy design-for-design (radar highest on both, small multiples lowest on both), this also supports treating F0 as a genuine check on whether participants engaged with the encoding's content rather than a design's overall shape memorability alone, as intended in Section 5.1.

Small multiples' weak performance across F0, F1, and M1 is notable given that, of the three fingerprint designs, it is the one whose reference view is the most visually distinctive at a glance (five separate maps, rather than five overlaid polygons or bar rows): distinctiveness at the reference-viewing stage evidently did not translate into distinguishability once the name was removed and the shape shown in isolation, which is itself informative about what makes a fingerprint encoding learnable, independent of how it happens to look side by side. All three designs nonetheless answered the P1 hardest-pair check well above the 50% chance baseline (77–85%), which suggests the underlying data supports pairwise distinguishability even where overall matching accuracy is weaker; the gap between M1 and P1 for small multiples in particular (49% vs 77%) indicates that the design's difficulty lies more in distinguishing all five patterns simultaneously than in any single pairwise comparison.

For the topology direction, the network and linear diagram matched each other on the single strongest-pair task (69% each) but diverged sharply on cluster recognition, with the linear diagram's 23% barely above what would be expected if participants selected among the three options at random (33%) and well below the network's 69%. Read together with the network's markedly stronger strength-rating correlation ($r = 0.64$ against $r = 0.30$), this points to the same underlying difference: encoding connection strength as edge thickness and spatial proximity (the network) communicates both a

single strongest pair and a broader multi-place cluster more reliably than encoding it as arc height along a single axis (the linear diagram), even though the two designs are statistically indistinguishable on the narrower single-pair task alone.

The metro map's result is the most interesting tension in the results. It performed worst of the three topology designs on the specific strongest-pair task (15%, against 69% for the other two) despite encoding exactly the same underlying weight as its line structure, yet it was decisively the most popular in the post-task ranking and the most frequently named "most memorable" design. A plausible reading, consistent with several participants' own explanations for their ranking choice ("similar to real transit maps I am familiar with", "its like a map that we see everyday"), is that the metro map's borrowed visual language is read fluently as a familiar transit map rather than as an instrument for locating one specific strongest edge among many; participants navigate it confidently, but confidence in navigating a representation is not the same as accuracy at the specific task this study asked of it. This is a genuine, if unintended, illustration of the same principle the design chapter draws from Beck's London Underground map (Section 3.5): the metro map optimises for a comfortable, generalisable mental model of the network as a whole rather than for precise retrieval of any single edge weight, which is exactly the trade-off the original inspiration for the design makes.

Cross-tabulating by expert/general-public group is reported here with the caveat, stated in Section 5.1 and repeated because it matters for how much weight to put on it, that four experts is too small a group for a robust quantitative comparison; a single participant's answer shifts the expert-group percentage by 25 points. Descriptively, the expert group's free-text answers were not obviously more accurate on the identification tasks, but were more likely to reference the underlying literary-geographic content directly rather than only the visual encoding, for example one expert participant's cross-design synthesis answer, "I was surprised that some places that are geographically quite separate still seem closely connected in the authors' writing. It made me realise that literary connections between places do not necessarily follow their physical locations," which restates the dissertation's own RQ2 argument in the participant's own words without having been told it in advance. Another expert participant's reflection, "Yes, Walter Scott in Leith I did not realise this is where his work was set," is a small but direct instance of the visualisation surfacing a literary fact the participant did not already know, which is closer to genuine spontaneous discovery than to guided instruction. Both quotes are treated as indicative rather than representative, given the sample size, but they are the kind of qualitative evidence the deep-insight questions (T3, T4, Cross-design

synthesis) were designed to surface, and their presence at all, unprompted, in a sample this small is worth weighing alongside the quantitative results above.

Chapter 6

Discussion

6.1 Design Improvements after User Study

The results in Section 5.4 point to concrete, design-specific changes rather than a single across-the-board fix, since the three fingerprint designs and three topology designs each failed, where they failed, in different ways.

For the small-multiples fingerprint design, the weakest performer on every objective measure, the gap between its own strong reference-view distinctiveness and its weak blind-identification accuracy suggests the problem is not legibility in the abstract but the loss of the side-by-side comparison the reference view provides: participants could tell five maps apart when shown together but struggled with one map in isolation. A design change worth testing is retaining a faint, unlabelled silhouette of the other four authors' spatial envelopes in the blind view, preserving the comparison structure without reintroducing any identifying label. For the bar-code design, the large Top-1-to-Top-2 accuracy gain (46% to 77%) suggests participants are extracting real signal (narrowing correctly to a plausible pair) without a reliable way to commit to a single answer; sorting bars by frequency rather than sector by default, rather than requiring participants to switch modes themselves, may make the single largest bar easier to commit to as the deciding feature, which several F3 answers already used as their stated strategy.

For the topology direction, the metro map's weak strongest-pair accuracy despite strong subjective preference is the clearest case for a targeted change: since the design's transit-map familiarity is a genuine strength worth keeping (Section 5.4), the more promising fix is adding a lightweight strength cue within that visual language, such as line thickness or a station-size encoding proportional to interchange weight, rather than asking participants to abandon the metro metaphor for a more data-forward one.

The linear diagram's comparatively poor cluster recognition (23%, barely above the three-option chance baseline) despite matching accuracy on the single-pair task suggests its single-axis arc layout communicates one connection at a time well but not the multi-place groupings a network or metro layout makes visually apparent through spatial adjacency; a targeted improvement would be arc-bundling or colour-coding arcs by community membership, giving the linear diagram an explicit cluster cue it currently lacks.

More generally, the F0 comprehension check's correlation with F1 identification accuracy across all three fingerprint designs (Section 5.4) suggests that future iterations of this evaluation could productively use F0-style objective legibility checks earlier in the design process itself, as a lightweight way to catch a design that looks distinctive but does not actually communicate its content, before committing to a full user study built around it.

6.2 Narrative Topology vs Geographic Topology

The most striking finding from the data analysis is the co-occurrence of Leith and Princes Street across 28 distinct literary works, the strongest co-occurrence pair in the entire corpus. Geographically the two places sit about 2.5 kilometres apart: Leith a port district historically associated with trade, working-class communities, and political radicalism, Princes Street the principal commercial artery of the Georgian New Town and bourgeois Edinburgh. On a geographic map they are simply two points in the same city. In the narrative topology visualisation they are the closest pair in the entire network.

That is not a coincidence. The journey between Leith and Princes Street recurs as a structuring device in Edinburgh fiction precisely because it crosses a social as well as a spatial boundary. Irvine Welsh's *Trainspotting* makes this explicit, with characters moving between Leith and the city centre as a movement between social worlds, between the margins and the mainstream of Edinburgh life; Walter Scott's historical novels stage similar journeys between the Old Town closes and the wider city as crossings between historical strata. A geographic map cannot show this. The narrative topology visualisation can: it shows these two places as *narratively adjacent*, regardless of their spatial distance.

This finding speaks directly to the theoretical framing of the project. Westphal's geocriticism argues that literary space is relational and constructed, not a reflection

of physical geography [11]. The Leith/Princes Street co-occurrence is a concrete, quantitative instance of this argument: the places that are most strongly connected in Edinburgh's literary imagination are not the places that are nearest to each other on a map.

The metro-map redesign (Section 3.5.0.0.1) provides a second, independent test of this claim, and a methodological one: if narrative topology genuinely diverges from geography, a line-grouping derived purely from co-occurrence weights should look different from one derived by hand along geographic axes, and should be checkable against the data in a way a hand-drawn grouping is not. The audit bore this out: only 22% of the hand-drawn map's same-line adjacencies were backed by real co-occurrence, against 88% once lines were derived from the co-occurrence graph itself, evidence in its own right that geographic intuition is an unreliable guide to narrative structure, consistent with Finding 4 rather than a separate result.

The redesign also surfaced two findings the geographic version could never have produced, since geographic grouping simply cannot detect them. Leith, considered as a place name across all five authors combined, groups with the historical Old Town corridor associated with Scott and Lockhart rather than with the contemporary, working-class Leith of Welsh's novels: at the level of the aggregated five-author corpus, the name's narrative associations pull more strongly toward nineteenth-century historical writing than toward the corpus's single most Leith-saturated author. That is a finding about how a place name functions across a corpus, not about any one author's usage of it, and it only becomes visible once co-occurrence, rather than an author's own usage, is what groups places together. Separately, a five-station corridor (Castle Street, Lasswade, the University of Edinburgh, the Royal Society of Edinburgh, and St Giles) emerged as a coherent cluster corresponding to Lockhart's biographical rather than fictional geography of Scott: these are places Scott lived and worked, as documented in *Life of Walter Scott*, not places his novels are set. The community-detection step distinguishes this biographical register from the historical-fiction register of the neighbouring Old Town corridor without being told to do so.

6.3 Author Spatial Languages

The near-total absence of shared place names across the five test authors (zero places shared by three or more, only 18 shared by any two) is a finding that can be stated as a simple number but deserves fuller interpretation. In effect, each author has constructed,

across their body of work, an Edinburgh that is almost entirely their own: McCall Smith's is bounded by the triangle formed by Scotland Street, Dundas Street, and Drummond Place in the New Town; Welsh's by Leith; Scott's is centred on the Canongate and the Old Town closes; Stevenson's extends south towards the Pentlands, reflecting his childhood at Swanston Cottage, and west towards the Firth of Forth, reflecting his father's lighthouse engineering work.

The radar chart and bar-code fingerprint make these private geographies comparable at a glance. A user who knows nothing of these authors' biographies can see, from the shape of the radar chart polygon, that McCall Smith and Welsh occupy different spatial territories within the same city. The visualisation encodes, in its geometry, a literary-geographic argument that would otherwise require paragraphs of prose.

This vocabulary separation also explains a non-obvious pattern observed when the network graph was scale-tested against the full corpus ([network_scale_explore.html](#)). Co-occurrence density does not fall smoothly as the number of authors grows: it is high both at the very small end (two authors, or a document set drawn from two or three books) and low at the very large end (all 408 authors), for two different reasons.

At the small end, document-level co-occurrence approaches its degenerate case, since almost any two places mentioned anywhere in the same handful of books will co-occur at least once, a single book's own place vocabulary trivially satisfying "appears in the same document." Restricting the network to three books (two by McCall Smith, one by Welsh) produced 152 places and 5,889 co-occurring pairs at weight ≥ 1 (51% of all possible pairs); raising the threshold to weight ≥ 2 collapsed this to 33 places and 424 pairs, almost all of them connecting McCall Smith's two books to each other rather than to Welsh's. That happens because his fictional Scotland Street recurs across his own works while Welsh's Leith does not recur in either of McCall Smith's, a direct, book-level instance of the vocabulary separation just described.

At the large end, the opposite problem appears: adding authors adds nodes far faster than it adds genuine cross-author edges, for the same underlying reason, so the 408-author graph is large (533 places, 23,380 pairs) but comparatively sparse (16.5% of possible pairs, against 79.6% for two authors). The five-author baseline used throughout this dissertation sits between these two failure modes, with enough authors for a cross-author edge to read as a meaningful, comparatively rare signal, rather than either the near-certainty of the two-author or two-book case or the near-absence of the full-corpus case.

6.4 Limitations and Open Questions

Corpus size confound. McCall Smith contributed 17 works to the corpus, Stevenson 16, Scott 12; Welsh only 8, Lockhart 6. Because sector distributions are normalised as proportions of each author’s total mentions, a large corpus and a small corpus with identical spatial habits would produce identical fingerprints, which is worth being upfront about. That same normalisation, though, means the results are best read as measures of *spatial attention* (what fraction of each author’s narrative geography goes to each sector) rather than absolute output, and the consistency of the spatial patterns across works within each author’s corpus, visible when individual books are examined, suggests these are genuine authorial preferences rather than an artefact of corpus size.

Five-author prototype, tested at scale. The current system uses five test authors, selected for contrast and representativeness. Rather than leave open whether this scales to the full corpus, it was tested directly for all six designs, using a set of scale-exploration tools kept separate from the six canonical five-author prototypes (radar, bar-code, network, and linear are live interactive pages that recompute their data client-side for any of the 408 authors with location data; small multiples and metro required batch or capped-live testing for reasons specific to each design, detailed below). The headline result is that the six designs do not share one scaling bottleneck. Each fails in a different, specific way, summarised in Table B.6 (Appendix B.6).

Radar and bar-code (Direction 1) both degrade by losing the “compare everything at a glance” property, but through different mechanics. Radar’s static polygon overlay becomes visually unreadable well before 408 authors, with individual shapes no longer distinguishable by 20, though a hover-to-isolate interaction recovers single-author legibility at any N at the cost of the all-at-once view. Bar-code never becomes unreadable or slow (408 rows render in 20 milliseconds), but the page ends up roughly 50 screens tall, so comparing two arbitrary authors means scrolling between points far apart rather than looking at two adjacent shapes.

Network and linear (Direction 2) share a root cause, one mark per co-occurring place pair laid out along a single spatial dimension with no wrapping, but it shows up differently in each. Network’s force-directed layout degrades gracefully: only the graph’s dense core becomes unreadable, and the periphery stays legible even at 408 authors. Linear’s fixed horizontal axis instead just grows the canvas to tens of thousands of pixels wide, so a screenshot of the visible, scrolled viewport looks almost unchanged from the five-author version even though the underlying canvas is roughly

27 screen-widths across.

Small multiples is the only design with a measured, genuine computational cost rather than a purely visual or navigational one, since each panel is a live map instance requesting real tiles. Initialisation time per panel stays roughly constant from 5 to 50 authors ($\approx 1\text{--}3\text{ms}$ each) but degrades by a factor of 3.6 from 50 to 408 (measured directly, not extrapolated, at 1.47 seconds total): a real main-thread delay a user would feel as a freeze, and distinct from the other designs’ purely visual degradation.

Metro is the only design where the failure sits in the underlying algorithm rather than the picture. Re-running the community-detection pipeline (Section 3.5.0.0.1) at larger N shows the adjacency-backed percentage declining monotonically (96% at 2 authors, 92% at 5, 84% at 20, 72% at 50), and at 50 authors one community grows to 75 stations because `greedy_modularity_communities` judges it internally cohesive enough that the recursive size cap cannot split it further. That is a structural limit of the clustering approach, not a rendering or layout problem.

Metro-map layout is heuristic, not optimal. The metro-map redesign (Section 3.5.0.0.1) replaces manual line assignment with community detection and manual station placement with a spring embedding corrected for octilinearity. Neither step solves the underlying schematic-layout problem exactly, since Nöllenburg and Wolff’s result [6] establishes that an exact solution is computationally hard in general, and two residual limitations follow directly from this.

First, the line structure is sensitive to the community-detection threshold and the five-author, top-15-places-per-author filter used to build the co-occurrence graph. This was confirmed directly rather than left as speculation (Appendix B.6): re-running the pipeline at 20 and 50 authors produces 8 and 11 lines respectively, with the proportion of adjacency-backed adjacent stations falling from 92% at 5 authors to 72% at 50, and at 50 authors one community grows to 75 stations because `greedy_modularity_communities` treats it as internally unsplittable despite exceeding the 18-station recursion cap. That is a structural limit of the clustering approach; the current five-line map avoids it only because it happens to sit at a favourable point on this curve, not because the underlying method guarantees it, so the current five lines should not be read as a fixed taxonomy. Second, the greedy label-placement search leaves two of 57 station labels overlapping in the single densest interchange cluster, which a fully general solver (or manual adjustment, common practice even for professionally drawn transit maps) could resolve.

PostGIS and polygon precision. The `api_location` table records places as latitude/longitude centroids, not polygons. For a place like “Old Town,” the centroid

may not accurately represent the full extent of the area referenced in the text. PostGIS polygon data is available in the original database schema but requires the PostGIS extension, which was not installed in this project. Future work could install PostGIS and use the polygon representations for more precise sector assignment.

Source text sampling. All six prototypes now surface example sentences drawn from `api_sentence` (Section 4), addressing the supervisor’s priority that readers should be able to see the literary sentence in which a place appears rather than only an aggregate count. The remaining limitation is that examples are sampled (with a fixed random seed, for reproducibility) rather than curated: a place with many mentions is shown a small, arbitrary subset of its sentences, which may not be the most illustrative or representative ones. A curated or relevance-ranked selection is left for the combined interface.

Reader-sketch and narrative time. The supervisor noted a direction labelled “Reader-sketch?” in early meetings, referring to visualisations that represent the reading experience: the sequence in which places are encountered as the reader moves through a text. The `position_pct` field in `mention_order` provides the necessary data for this, and Reader-plot timeline visualisations were prototyped but not included in the user study. This remains an open direction.

User study sample size. The evaluation reported in Chapter 5 draws on $n = 13$ participants, below the 13–20 planned during recruitment and well below what would support statistically powered comparisons; percentages throughout Chapter 5 should be read as directional rather than precise estimates, with a single participant’s answer worth roughly 8 percentage points on any given measure. The domain-expert subgroup ($n = 4$) is thinner still, and the expert/general-public comparisons in Section 5.4 are accordingly treated as indicative rather than conclusive throughout. Recruitment was also convenience-based, through the researcher’s own personal and academic network rather than a broader public call, which likely skews the sample toward participants already favourably disposed to engage carefully with the task; this is a plausible contributor to the generally low rates of self-reported prior confusion with the visualisations (Q-R3), and a larger, more broadly recruited sample would be needed before drawing firm conclusions about how well these designs generalise beyond a personal network. The exclusion of three low-quality responses (Section 5.3) was based on researcher judgement against criteria (completion time, free-text engagement) that were not formally pre-registered before data collection began, which is a standard limitation of a small, time-constrained study rather than one specific to this project, but is stated here for transparency.

Chapter 7

Conclusion

This dissertation has argued that place names in literary text are data about narrative structure rather than merely geographic location, and has shown that this argument can be operationalised as a set of interactive visualisations grounded in real corpus data. The LitLong Edinburgh database, 620 literary works spanning three centuries, 2,135 place names, 50,248 mention records, provided the empirical foundation, and six visualisation prototypes were developed across two research directions, implemented in D3.js, and evaluated through a user study.

RQ2 — does narrative co-occurrence reveal a topology distinct from geography? Yes. The strongest evidence is the Leith/Princes Street pair itself: 28 shared works despite roughly 2.5km of geographic separation, and one of only 12 station-pairs across the entire redesigned metro map that behave as a genuine narrative interchange. The redesign of the metro-style map (Section 3.5.0.0.1) turned this from a single anecdote into a systematic result, deriving line structure from measured co-occurrence rather than hand-drawn geography and raising the proportion of adjacent stations sharing a real book from 22% to 88%, while surfacing two further instances of the same divergence that geography could not have produced: the word “Leith” clustering with Scott and Lockhart’s historical corridor rather than Welsh’s, and a biographical corridor (Castle Street, Lasswade, the University of Edinburgh) distinct from the historical-fiction corridor immediately next to it. The scale-exploration work (Appendix B.6) adds a boundary condition here rather than a contradiction: narrative and geographic topology diverge clearly at the five-author scale used throughout this dissertation, but the mechanism generating that divergence, near-total non-overlap of authorial vocabularies (Finding 1), also means the co-occurrence signal thins out at very large author counts and degenerates at very small ones (2–3 books). The divergence is most legible at a middle scale, in

other words, rather than a universal property independent of N .

RQ1 — does an author’s spatial distribution form a visually distinctive, learnable fingerprint? Partially yes. Across the user study ($n = 13$), the radar chart supported the strongest evidence for this claim: after studying all five authors’ fingerprints together, participants correctly matched all five unlabelled shapes back to their authors at 69% accuracy, well above the 20% chance baseline, and correctly distinguished the single most similar pair of authors 85% of the time. The bar-code and small-multiples designs showed the same direction of effect but more weakly, with matching accuracy close to half (48% and 49% respectively) rather than clearly distinguishable, and single-shape identification in isolation (23–46% across the three designs) was consistently harder than matching all five side by side, indicating that these fingerprints are more reliably told apart from one another than recognised from memory alone. The evidence therefore supports RQ1 for at least one encoding, the radar chart, without supporting a stronger claim that spatial fingerprinting works equally well regardless of how the fingerprint is drawn; which specific encoding choices explain the radar chart’s advantage, and whether it would hold at a larger sample size, is discussed in Chapter 6.

The project’s contributions include: a validated spatial classification of Edinburgh place names using official City of Edinburgh Council administrative boundary data; an empirical finding that each of five representative Edinburgh authors constructs an almost entirely private imaginary geography of the city; a method for extracting narrative reading order from the LitLong database using the `start_word` field as a monotonic index; and a suite of interactive visualisation prototypes that make authorial spatial patterns and narrative co-occurrence structure visible in forms not previously available.

Several directions for future work follow from where the project currently stands. The combined interface (Section 4.5) already provides author-selection controls reaching the full 408-author corpus for five of the six views, with rendering safety caps keeping it responsive at scale, but the metro map’s dependence on offline community detection is still a gap: it currently limits the map to a small number of precomputed author-count snapshots rather than an arbitrary live selection, something a faster, approximate community-detection method running client-side could close. The interactive detail panels are likewise already grounded in `api_sentence` source text (Section 4.5), and a natural extension would be a single outer panel shared consistently across all six canonical views, rather than each view’s own panel plus the combined interface’s separate one. Beyond the interface itself, the `api_posmention` table’s two million-plus part-of-speech annotations would support a further research direction in narrative

weight analysis: whether a place name appears in dialogue, narration, or description, and whether this varies systematically across authors or genres. And the concept of “literary silences,” Edinburgh places that appear in the gazetteer but never in any literary work, would complement the current analysis by revealing what Edinburgh’s literary imagination systematically excludes.

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Appendix A

Study Material

A.1 Participant Information Sheet

Participant Information Sheet

Project title:	Studying Experiences with Data Representations of a Collection of Edinburgh-related Literature
Principal investigator:	Uta Hinrichs
Researcher collecting data:	Merritt Wang
Funder (if applicable):	N/A

This study was certified according to the Informatics Research Ethics Process, reference number 446635. Please take time to read the following information carefully. You should keep this page for your records.

Who are the researchers?

This research conducted by Merritt Wang as part of their MSc project at the School of Informatics, University of Edinburgh. The MSc project is supervised by Dr. Uta Hinrichs, Reader in Data Visualization at the School of Informatics.

What is the purpose of the study?

This MSc project focuses utilizing data representations (e.g., data visualizations or physicalizations to make a collection of Edinburgh-related literature explorable from different perspectives. As part of this study we would like to explore people's interest in the data and their experiences of bespoke data representation.

Why have I been asked to take part?

You have been asked to take part in this study as you live in Edinburgh, have indicated an interest in Edinburgh-related literature, and/or have expressed an interest in this project. Your participation in this study will help us design and refine relevant data representations to make this collection of Edinburgh-related literature explorable by a wide range of audiences.

Do I have to take part?

No – participation in this study is entirely up to you. You can withdraw from the study at any time, up until **7 days after your study session** without giving a reason. After this point, personal data will be deleted and anonymised data will be combined such



that it is impossible to remove individual information from the analysis. Your rights will not be affected. If you wish to withdraw, contact the PI – Dr. Uta Hinrichs (uhinrich@ed.ac.uk). We will keep copies of your original consent, and of your withdrawal request.

What will happen if I decide to take part?

You will first be asked to complete a short background questionnaire about your interest in or experience with Edinburgh-related literature and/or data visualisation (approx. 5 min.). You will then be introduced to several data visualisations representing Edinburgh-related literature and asked to complete a small number of tasks based on them (for example, identifying which author a visualisation belongs to, or judging the relationship between places), followed by a short reflection questionnaire about your experience. Your task responses and questionnaire answers will be recorded in writing; no audio or video recording will take place. The full session should take approx. 35-50 min. to complete.

Are there any risks associated with taking part?

There are no significant risks associated with participation.

Are there any benefits associated with taking part?

There are no direct benefits for participating in these study activities, other than learning about Edinburgh-related literature and getting the opportunity of informing ways of exploring it.

Refreshments will be provided for in-person study activities.

What will happen to the results of this study?

The results of this study may be summarised in published articles, reports and presentations. Quotes or key findings will be anonymized: We will remove any information that could, in our assessment, allow anyone to identify you. With your consent, information can also be used for future research. Your data may be archived for a maximum of 4 years. All potentially identifiable data will be deleted within this timeframe if it has not already been deleted as part of anonymization.



Data protection and confidentiality.

Your data will be processed in accordance with Data Protection Law. All information collected about you will be kept strictly confidential. Your data will be referred to by a unique participant number rather than by name. Your data will only be viewed by the researcher/research team named on this PIS.

All electronic data will be stored on a password-protected encrypted computer, on the School of Informatics' secure file servers, or on the University's secure encrypted cloud storage services (DataShare, ownCloud, or Sharepoint) and all paper records will be stored in a locked filing cabinet in the PI's office. Your consent information will be kept separately from your responses in order to minimise risk.

What are my data protection rights?

The University of Edinburgh is a Data Controller for the information you provide. You have the right to access information held about you. Your right of access can be exercised in accordance Data Protection Law. You also have other rights including rights of correction, erasure and objection. For more details, including the right to lodge a complaint with the Information Commissioner's Office, please visit www.ico.org.uk. Questions, comments and requests about your personal data can also be sent to the University Data Protection Officer at dpo@ed.ac.uk.

Who can I contact?

If you have any further questions about the study, please contact the lead researcher, Uta Hinrichs (uhinrich@ed.ac.uk).

If you wish to make a complaint about the study, please contact inf-ethics@inf.ed.ac.uk. When you contact us, please provide the study title and detail the nature of your complaint.

Alternative formats.

To request this document in an alternative format, such as large print or on coloured paper, please contact Uta Hinrichs (uhinrich@ed.ac.uk).

General information.



For general information about how we use your data, go to: [Privacy notice |
Edinburgh Research Office | Edinburgh Research Office](#)



A.2 Consent Form

Participant number: _____

Participant Consent Form

Project title:	Studying Experiences with Data Representations of a Collection of Edinburgh-related Literature
Principal investigator (PI):	Uta Hinrichs
Researcher:	Merritt Wang
PI contact details:	uhinrich@ed.ac.uk

By participating in the study you agree to take part in the study activities listed in the Participant Information Sheet.

- I have read and understood the Participant Information Sheet for the above study, that I have had the opportunity to ask questions, and that any questions I had were answered to my satisfaction.
- My participation is voluntary, and that I can withdraw at any time without giving a reason. Withdrawing will not affect any of my rights.
- I consent to my anonymised data being used in academic publications and presentations.
- I understand that my anonymised data will be stored for the duration outlined in the Participant Information Sheet.

Please tick yes or no for each of these statements.

1. I allow my data to be used in future ethically approved research.

☐	☐
Yes	No

2. I agree to take part in this study.

☐	☐
Yes	No

Name of person giving consent

Date
dd/mm/yy

Signature

Name of person taking consent

Date
dd/mm/yy

Signature



A.3 Task Questions

The following instrument implements the two task types described in the Study Design section of Chapter 5. Each participant sees all six visualisations; the order of the three fingerprint designs (and, separately, the three topology designs) is randomised per participant to control for ordering effects. Wording is final; formatting into the online survey tool is the only remaining step before the study can run.

Task 1 — fingerprint identification and matching (shown once per fingerprint design: radar, bar-code, small multiples). Participants first see a reference view, the canonical all-five-authors version of the design in question, and are asked to spend a few minutes noticing how the five shapes differ before continuing; no hint is given as to what to look for. They then see a single-author variant of the same design, showing one author's shape with no legend or name anywhere, and answer:

1. Based on the chart shown above (not your memory of the reference view), which sector/place does this author's writing concentrate on the most? (*single choice among four options: the true answer plus the three next-highest sectors/places for that author by real value, so the distractors are genuinely close rather than arbitrary*)
2. You are now looking at one of the five fingerprints you just saw, with the name hidden. Which of the following five authors do you think it represents? (*Alexander McCall Smith / Irvine Welsh / John Gibson Lockhart / Walter Scott / Robert Louis Stevenson / I don't know*)
3. If your first guess turned out to be wrong, which of the five would be your second guess? (*same five authors, forced choice*)
4. How confident are you in your first answer? (*Not at all confident – Very confident, 5-point scale*)
5. What specific feature of the shape led you to that answer (for example, an unusually large spike in one area, or a shape that is spread evenly across many areas)?
6. If no one had told you anything about this visualisation, would you have understood how to read it just by looking? (*Yes, easily / Yes, with a little effort / No, I was confused*)

A single forced-choice identification (item 2) only ever yields a correct/incorrect judgement against a one-in-five baseline; the runner-up question (item 3) is added cheaply, from the same stimulus, to distinguish participants who narrowed the shape down to a plausible pair from those guessing entirely at random, since both would otherwise register identically as “incorrect.” Item 1 addresses a different concern from items 2–6: even with the reference-view colour removed from the blind stimulus (Chapter 5), a participant could in principle answer item 2 correctly by matching the shape’s overall silhouette against a memorised image from the reference view, without the shape’s specific content – which sector or place it actually peaks toward – playing any role in that judgement, which would test recognition memory for an outline rather than comprehension of what the encoding communicates. Item 1 is answerable from the blind stimulus alone, with no memory of the reference view required at all, and is scored against a real answer key (the true highest-value sector or place for that author, against the next three highest as distractors) rather than the identification question in item 2, making it a direct, memorisation-resistant check of whether the encoding’s actual content was read rather than its shape merely recognised.

Finally, participants see a third variant of the same design showing all five authors’ shapes again, side by side, with no names and in a shuffled order distinct from the order used in the reference view, and answer:

7. Match each of the five unlabelled shapes to an author. (*matrix-table question: one row per shape, the same five authors as answer options in every row; a participant may assign the same author to more than one shape*)
8. Overall, having compared all five side by side, how easy or difficult do you think it would be to mix up any two of these five patterns? (*Very easy to confuse – Very easy to tell apart, 5-point scale*)
9. Look again at the two specific shapes identified (across all authors for this design) as most similar to one another by cosine similarity on their underlying value vectors. One is [Author A] and the other is [Author B]. Which is which? (*forced choice between the two orderings*)

The single-item identification question (item 2) yields only a correct/incorrect judgement against a one-in-five baseline; the matching question (item 7) instead yields a full 5×5 confusion matrix per design, indicating not just whether a design is distinguishable overall but which specific pairs of authors’ fingerprints are most often mistaken for one

another – a substantially richer test of the same RQ1 success criterion. The subjective rating (item 8) is compared against the matching question’s objective accuracy as a check for over- or under-confidence in participants’ own sense of how distinguishable the shapes are. Item 9 addresses a risk specific to item 7: if a design’s five shapes happen to be very easy to tell apart overall, the confusion matrix can saturate near-perfect and say nothing about relative distinguishability between designs. Item 9 targets the single most confusable pair of authors for that design specifically – for the radar chart, John Gibson Lockhart and Walter Scott (cosine similarity 0.844 on their 14-sector value vectors, the highest of all ten author pairs); for the bar-code, Walter Scott and Irvine Welsh (the second-highest pair for that design, chosen instead of the top pair to avoid repeating the same two authors’ names across two of the three fingerprint designs); for small multiples, John Gibson Lockhart and Robert Louis Stevenson (highest pair by cosine similarity on sector-aggregated mention counts) – guaranteeing a difficulty floor that a well-separated overall matrix cannot obscure, while the two-alternative format keeps chance performance at a known 50% rather than the one-in-five baseline of item 2. As with the single-author variant, the assignment of shapes to shuffled positions in the matching view is fixed rather than randomised per participant, since the matching stimulus is hosted outside the survey tool and there is no channel to record which shuffle a given participant saw; a different fixed shuffle is used for each of the three designs to prevent a participant from carrying a positional guess across designs.

Task 2 — narrative/geographic relationship (shown once per topology design: force-directed network, linear connection diagram, metro map). Participants are first invited to explore the visualisation freely (hover, click, zoom/scroll as the design allows) before answering, rather than being directed straight to the strongest pair; this is intended to surface spontaneous observations about the co-occurrence structure beyond the single forced-choice item, since a single near/far judgement on one pair alone would be too narrow a test of whether participants independently perceive the narrative-topology structure.

1. Looking at this visualisation, which two places appear to be the *most strongly connected*? (*free text, two place names*)
2. What about this diagram made you think these two places are the most connected (for example: the thickest line between them, how close together they are drawn, being on the same coloured line)? (*free text*)
3. Based on your own knowledge of Edinburgh (or your best guess if you are not

familiar with the city), are these two places geographically close together or far apart? (*Very close – Very far, 5-point scale / Not sure*)

4. For each of three specific place pairs, how strongly connected do they appear in this visualisation? (*matrix-table/rating-grid question, one row per pair, 5-point scale: Not connected at all – Extremely strongly connected*)
5. Beyond that pair, is there any *other* connection in this visualisation that surprises you — for example, two places that seem strongly linked despite seeming unrelated, or two places you would expect to be linked that do not appear connected? (*free text: place names + brief explanation, optional but encouraged*)
6. Do you notice any group of three or more places that seem to form a cluster of connections with each other? If so, list them and, if you have a guess, why you think they might be grouped together. (*free text, optional*)
7. Which of three candidate groupings of places looks like it forms the tightest cluster in this visualisation? (*single choice among three options, order randomised; one option is a genuine cluster, the other two are constructed by mixing places from different clusters*)
8. If no one had told you anything about this visualisation, would you have understood how to read it just by looking? (*Yes, easily / Yes, with a little effort / No, I was confused*)

Items 4 and 7 are added for the same reason as the fingerprint task's item 8 above: item 1 alone risks a ceiling effect if the strongest pair in a given dataset is visually unmistakable (here, Leith and Princes Street, at a co-occurrence weight of 28 against a next-highest weight of 18), and item 6 alone risks a floor effect if participants who notice nothing simply skip it, yielding no usable data either way. Item 4 asks participants to rate three place pairs deliberately chosen to span strong, medium, and weak ground-truth co-occurrence weight, rather than search for a pair themselves, producing interval-scaled data that can be correlated against the real weights regardless of whether item 1 saturates. Item 7 gives a forced choice between one real cluster and two distractors built by mixing places from different clusters, providing a scoreable recognition-based measure to sit alongside item 6's open-ended production task; agreement between what a participant names unprompted in item 6 and what they select in item 7 is treated as corroborating evidence in the analysis.

Item 2 addresses a separate risk specific to this task direction: the force-directed network, linear diagram, and metro map are three renderings of the *same* underlying co-occurrence weights, so the ground-truth answer to item 1 (and, before this revision, to items 4 and 7) is identical across all three designs. A participant who solves item 1 correctly on the first topology design they see – whether through genuine reading of that design or simply from recalling that Leith and Princes Street are heavily co-mentioned in Scottish literature – could reuse the same answer on the other two designs without engaging with either rendering at all, which would test recall of a previous answer rather than each design’s own capacity to communicate connection strength. Item 2 cannot be scored automatically, but a participant’s justification is expected to reference that specific design’s visual encoding (edge thickness or node proximity for the force-directed network, arc height for the linear diagram, shared-line adjacency for the metro map); an answer that is generic, absent, or inconsistent with the encoding actually used is treated during analysis as evidence the corresponding item 1 answer may have been carried over rather than freshly determined. Items 4 and 7 are also now varied across the three topology designs rather than reusing identical place pairs and cluster options: for the force-directed network, the three rated pairs are New Town–Princes Street (weight 18), Lochend–Waverley Station (weight 4), and Leith–Silvermills (weight 2), and the verified cluster is New Town, Princes Street, Dundas Street, and Stockbridge (one Louvain community on the shared co-occurrence graph); for the linear diagram, the pairs are Dundas Street–Princes Street (17), Old Town–Princes Street (6), and Arthur’s Seat–Haddington (2), and the verified cluster is Leith Walk, Lochend, Pilrig, and Waverley Station (a second, distinct Louvain community on the same graph); for the metro map, the pairs are Bruntsfield–Dundas Street (16), Howe Street–Stockbridge (6), and Dalkeith–Linlithgow (2), and the verified cluster is drawn from that design’s own author-named lines (the stations of “Lockhart’s Edinburgh”). A participant therefore has to locate a new set of places in whichever rendering is currently in front of them rather than reuse a rating or selection made on an earlier design.

Cross-design synthesis (shown once, after all three topology designs have been seen).

1. Having now seen all three ways of showing connections between places, is there anything about Edinburgh’s literary geography that surprised you, or that you had not thought about before — for example, places that seem to “belong together” in these authors’ writing that you would not have expected? (*free text, optional but encouraged*)

Post-task ranking (once per direction, after all three designs in that direction have been seen).

1. Of the three fingerprint designs you just saw (radar chart, bar-code, small multiples), which did you find the most intuitive to read without any explanation? *(single choice + free-text reason, optional)*
2. Of the three topology designs you just saw (force-directed network, linear connection diagram, metro map), which did you find the most intuitive to read without any explanation? *(single choice + free-text reason, optional)*

A.4 Survey Questions

Background (before the tasks). Collected to test whether familiarity moderates the results reported in Chapter 5, not for identification purposes, consistent with the ethics application’s data-minimisation commitment.

1. How familiar are you with Edinburgh’s literary history (e.g. Scott, Stevenson, contemporary Edinburgh fiction)? *(Not at all – Very familiar, 5-point scale)*
2. Have you read any works by Alexander McCall Smith, Irvine Welsh, John Gibson Lockhart, Walter Scott, or Robert Louis Stevenson? *(Yes, several / Yes, one or two / No)*
3. How would you describe your background? *(Literary studies / digital humanities / information visualisation or HCI / other academic / general public with no specialist background / other, please specify)* — this question, not a fixed group assignment, is what actually separates the “domain expert” and “general public” groups described in Chapter 5 at analysis time.
4. How familiar are you with reading data visualisations or charts in general? *(Not at all – Very familiar, 5-point scale)*
5. Have you lived in or spent significant time in Edinburgh? *(Yes, I live/have lived there / I’ve visited several times / I’ve visited once or twice / Never)* — included to separate familiarity with the city itself from familiarity with its literature, which the corpus analysis (Chapter 3) treats as distinct.

Reflection (after all tasks).

1. Before this study, had you thought about literary place names as anything other than locations on a map?
2. Did any of the six visualisations change how you think about the relationship between a literary work and the real city it is set in? If so, which one, and how?
3. Is there anything about any of the six visualisations that confused you or that you would change?
4. Any other comments?
5. Overall, which single visualisation of all six did you find the most memorable or interesting, and why? (*free text*)

Appendix B

Data and Implementation

B.1 Database Schema

The LitLong Edinburgh PostgreSQL database contains 27 tables after full import. The tables most relevant to this project are listed in Table B.1.

Table B.1: Key tables in the LitLong Edinburgh database.

Table	Rows	Description
api_document	620	Literary works (title, publication date, type)
api_author	424	Authors (name, gender, Open Library ID)
api_document_author	1,232	Many-to-many: documents ↔ authors
api_location	2,135	Place names (text, latitude, longitude)
api_locationmention	50,248	Mention records (start_word, page_id, sentence_id)
api_sentence	50,248	Full source text for each mention
api_genre	29	Literary genre classifications
api_posmention	2,085,834	Part-of-speech annotations
mention_order	50,248	Custom: narrative order and position (derived)

B.2 Sector Definitions

Table B.2 lists the 14 sectors used in the spatial classification, their data sources, and their place-name counts.

Note: Nearest-neighbour counts are included in the exact-match totals above. Source: City of Edinburgh Council Open Spatial Data Portal (Open Government Licence v3.0). “Copyright

Table B.2: The 14 spatial sectors and their place-name counts.

Sector	Source	Exact matches	Nearest-neighbour
Old Town	Natural Neighbourhoods	296	—
South Central	NP Areas	255	—
New Town	Natural Neighbourhoods	217	—
Leith	NP Areas	152	—
Inverleith	NP Areas	140	—
Craighentenny/Duddingston	NP Areas	73	—
Pentlands	NP Areas	66	—
Canongate	Natural Neighbourhoods	64	—
Almond	NP Areas	64	—
South West	NP Areas	44	—
Western Edinburgh	NP Areas	43	—
Portobello/Craigmillar	NP Areas	38	—
Liberton/Gilmerton	NP Areas	36	—
Forth	NP Areas	27	—
Outer Scotland (excluded)	—	232	—

City of Edinburgh Council, contains Ordnance Survey data © Crown copyright.”

B.3 Author Selection

The five test authors were selected on the basis of mention count, number of works in the corpus, and thematic contrast. Table B.3 summarises their corpus statistics and dominant sectors.

Table B.3: Five test authors and their corpus statistics.

Author	Mentions	Works	Dominant sector	%
Alexander McCall Smith	7,320	17	New Town	43.6
Irvine Welsh	2,634	8	Leith	44.3
John Gibson Lockhart	3,046	6	New Town	25.9
Walter Scott	2,796	12	Old Town	32.0
Robert Louis Stevenson	1,778	16	Old Town	20.7

B.4 Co-occurrence Analysis

Table B.4 lists the ten strongest document-level co-occurrence pairs in the corpus, computed across the five test authors.

Table B.4: Top 10 document-level co-occurrence pairs (five test authors).

Place A	Place B	Weight (documents)
Leith	Princes Street	28
New Town	Princes Street	18
Dundas Street	Princes Street	17
Bruntsfield	Dundas Street	16
Bruntsfield	Princes Street	16
Dundas Street	New Town	16
Dundas Street	Morningside	15
Morningside	Stockbridge	15
Princes Street	Stockbridge	15
Dundas Street	The Meadows	15

B.5 Metro-Map Line Composition

Table B.5 lists the five lines produced by the community-detection pipeline described in Section 3.5.0.0.1, with the number of home stations (stations whose narrative cluster this line is), the total station count including spliced-in interchange stops, total mentions summed across home stations, and the adjacency-backed diagnostic (the number of home-station-adjacent pairs, out of the total possible, that share at least one book in common).

Table B.5: The five metro-map lines and their composition.

Line	Home stations	Total stops	Mentions	Adjacency backed
Smith's Edinburgh	15	16	2,786	13/14 (93%)
Leith & Forth Corridor	12	23	1,564	9/11 (82%)
Scott's Edinburgh	13	14	826	9/12 (75%)
Welsh's Edinburgh	12	12	494	11/11 (100%)
Lockhart's Edinburgh	5	5	202	4/4 (100%)
Total	57	—	5,872	46/52 (88%)

Note: “Total stops” exceeds “home stations” where a line has been joined by an interchange station whose home cluster is a different line (12 stations are interchanges across the whole map). The 22% adjacency-backed figure for the earlier, hand-drawn eight-line version was computed over 67 same-line adjacent pairs, 15 of which shared a book.

B.6 Scale Exploration Summary

All six designs were re-tested at four benchmark author counts (2, 5, 20, and either 408 or the largest value practical for the design in question), plus a 2–3 book test for network and linear, using scale-exploration tools kept separate from the six canonical five-author prototypes (`data/processed/scale_exploration/`). Table B.6 summarises the failure mode identified for each design; the full quantitative detail (per- N tables, timing measurements, and the decision log explaining every scope choice) is recorded in `data/processed/scale_exploration/NOTES.md` rather than reproduced in full here.

Table B.6: Scaling failure mode identified for each of the six designs.

Design	Failure mode	Key evidence
Radar	Visual overlap	Polygons indistinguishable by $N=20$; unreadable at 408; hover-to-isolate recovers single-author legibility at any N .
Bar-code	Navigation cost	408 rows render in 20ms (no rendering problem) but the page is $\approx 45,000$ px tall, about 50 screens of scrolling.
Network	Visual overlap (graceful)	Core becomes a dense mass at 408 authors; periphery stays legible. Density is U-shaped in N (79.6% at 2 authors, $\approx 25\%$ at 5, 16.5% at 408), driven by vocabulary non-overlap (Finding 1), not raw node count.
Linear	Horizontal sprawl	Canvas width grows from 1,100px (5 authors) to 29,790px (408 authors); a screenshot of the scrolled viewport looks unchanged because the same high-mention places sort leftmost regardless of N .
Small multiples	Computational cost (the only one)	Live Leaflet map initialisation degrades $3.6\times$ per-panel from 50 to 408 authors (1.47s total at 408, measured directly); the one design where scale risks a real freeze, not just illegibility.
Metro	Clustering algorithm breakdown	Adjacency-backed proportion falls monotonically ($96\% \rightarrow 72\%$, $N=2$ to 50); at $N=50$ one community reaches 75 stations because modularity detection judges it unsplittable despite exceeding the 18-station cap.